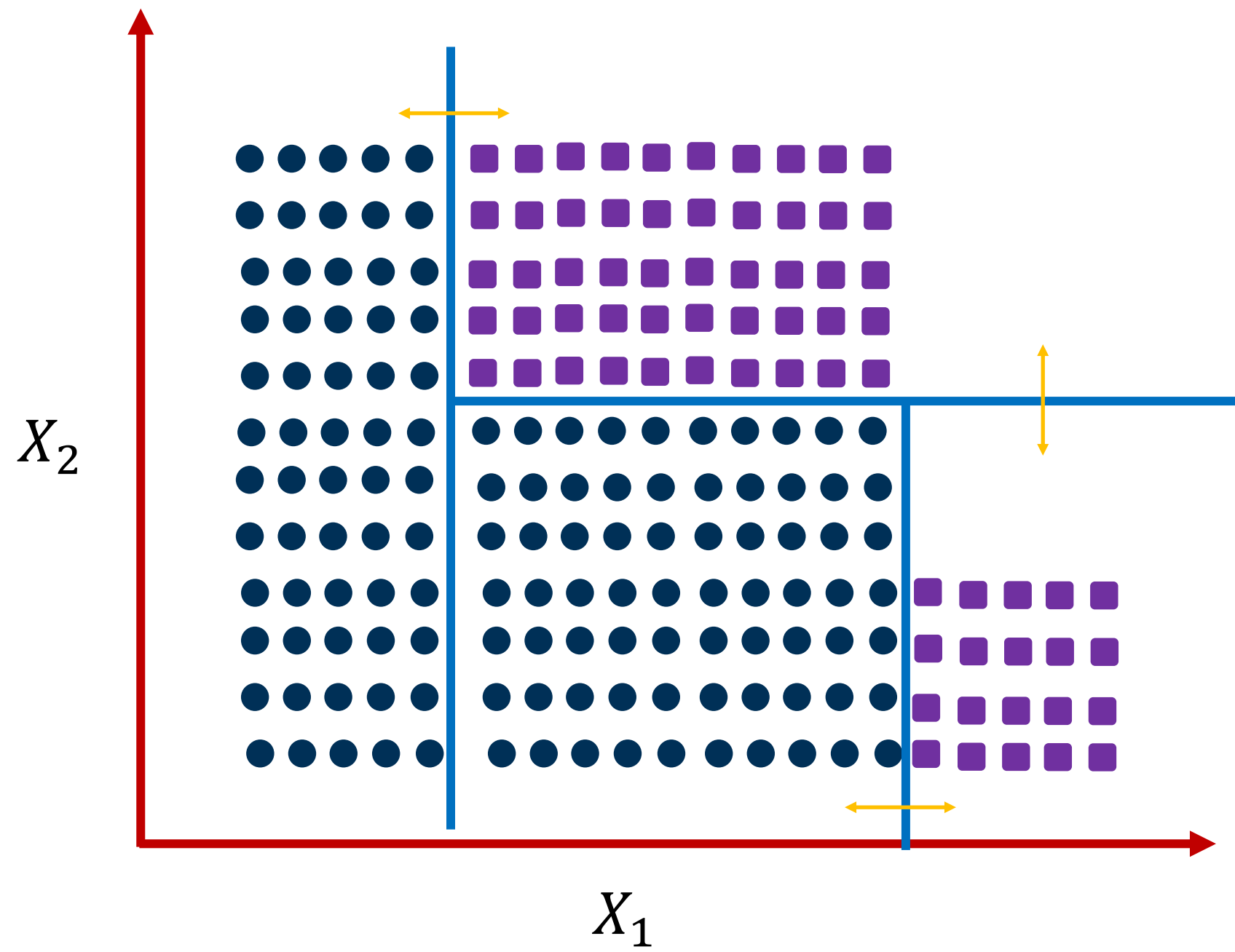
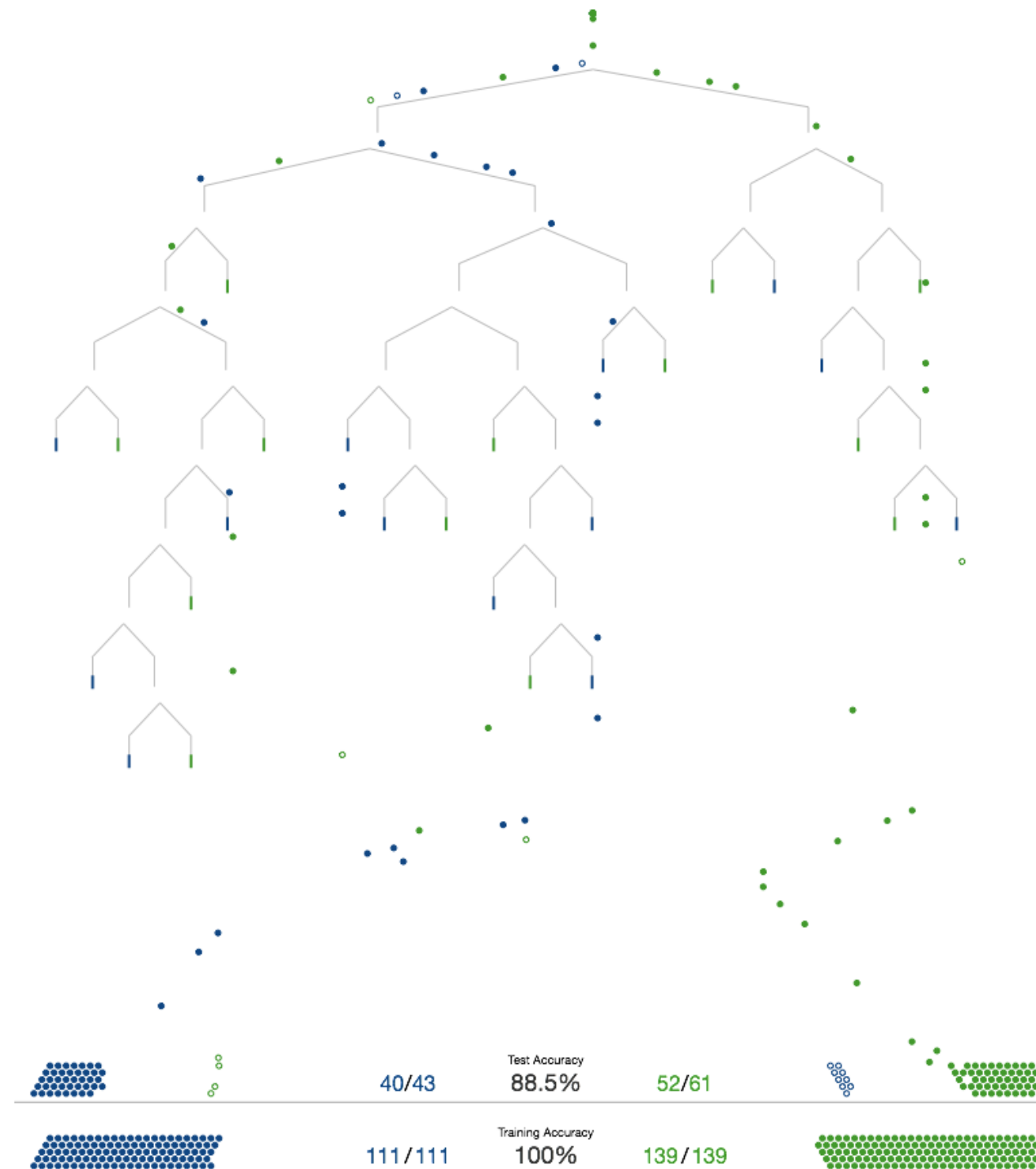


Decision Tree

Nimisha Roy
Georgia Tech



Visual Introduction to Decision Tree



Building a tree to distinguish homes in New York from homes in San Francisco

Decision Tree: Example (2)

Categorical Variables

feature dimension
attribute

	O	T	H	W	Play?
1	S	H	H	W	-
2	S	H	H	S	-
3	O	H	H	W	+
4	R	M	H	W	+
5	R	C	N	W	+
6	R	C	N	S	-
7	O	C	N	S	+
8	S	M	H	W	-
9	S	C	N	W	+
10	R	M	N	W	+
11	S	M	N	S	+
12	O	M	H	S	+
13	O	H	N	W	+
14	R	M	H	S	-

Outlook:

Sunny,
Overcast,
Rainy

Temperature:

Hot,
Medium,
Cool

Humidity:

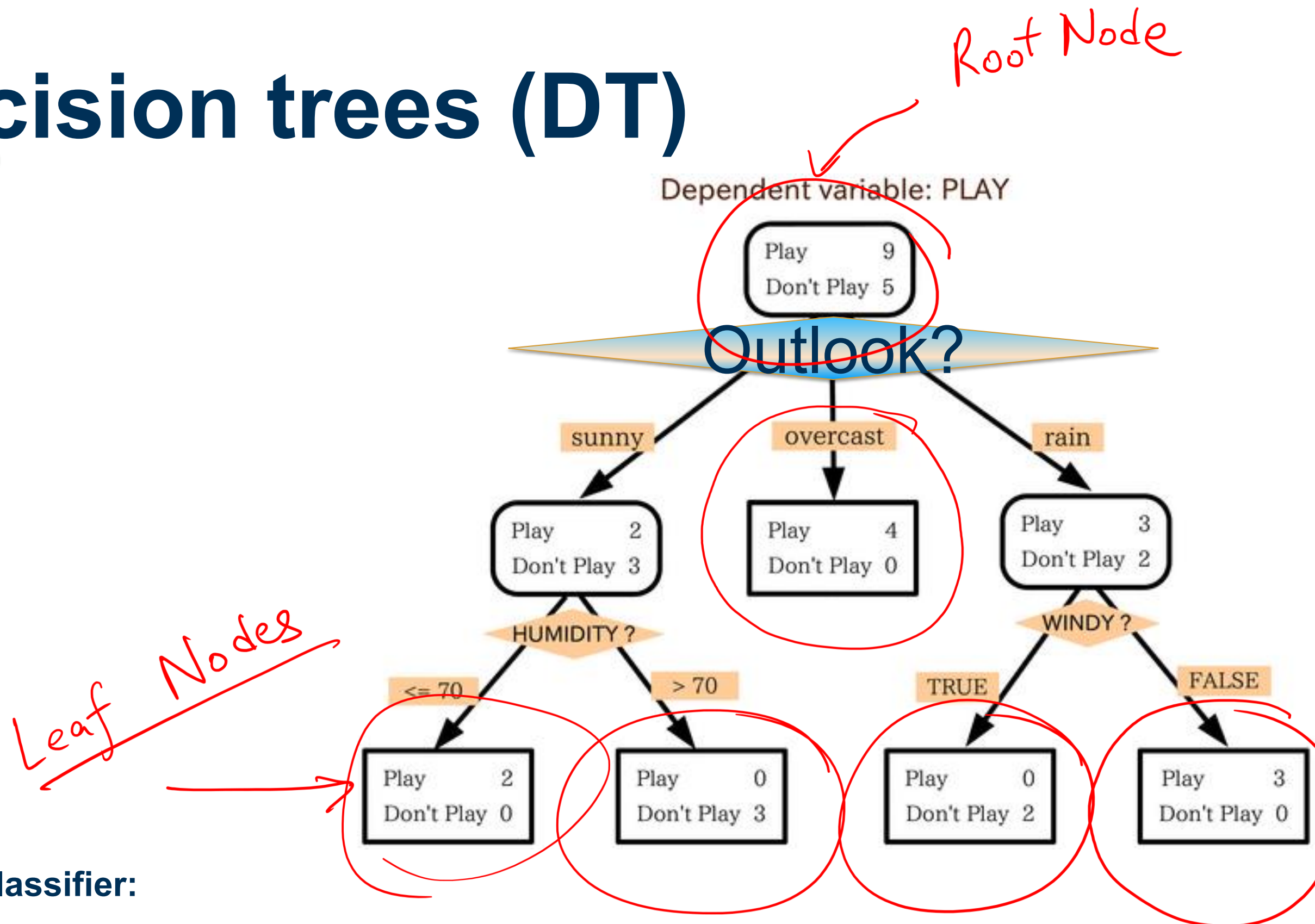
High,
Normal,
Low

Wind:

Strong,
Wweak

Will I play tennis today?

Decision trees (DT)



The classifier:

$f_T(x)$: majority class in the leaf in the tree T containing x

Model parameters: The tree structure and size

Decision trees

Pieces:

1. Find the **best attribute** to split on
2. Find the **best split** on the **chosen attribute**
3. Decide on when to stop splitting

← Possibly Overfitting

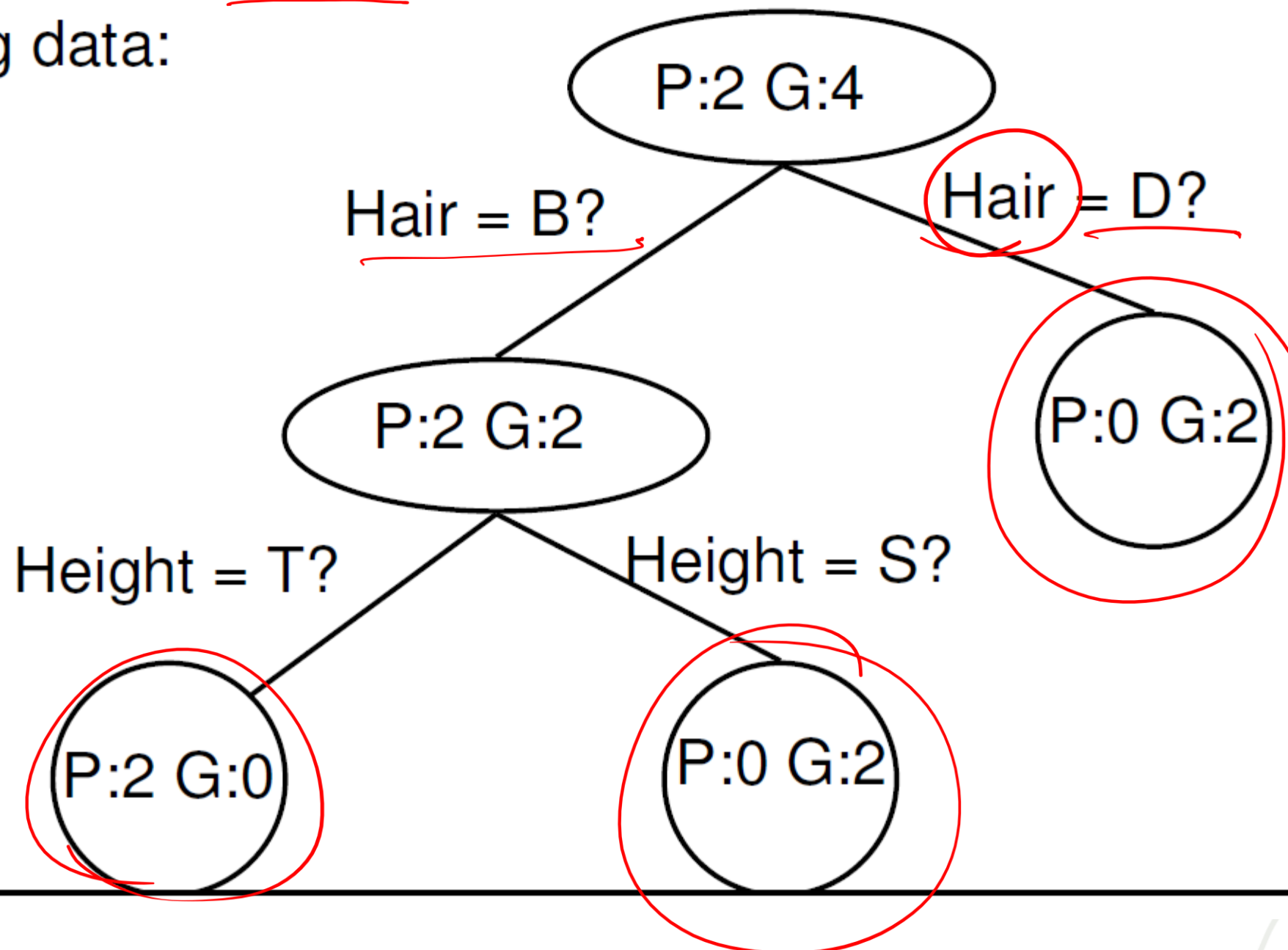
Categorical or Discrete attributes

- Three variables:
 - Hair = {blond, dark}
 - Height = {tall, short}

Label – Country = {Gromland, Polvia}

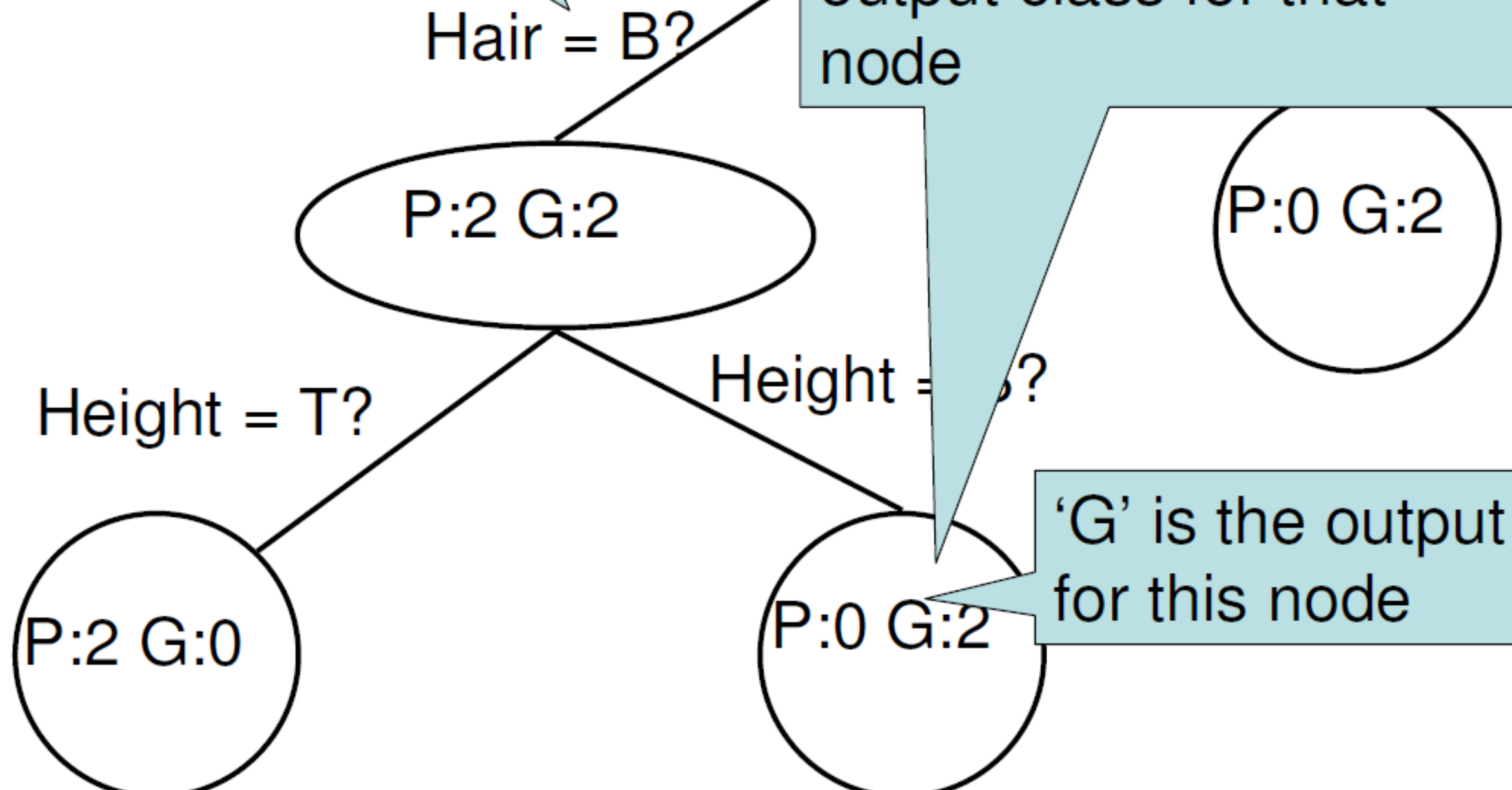
Training data:

(B,T,P)
(B,T,P)
(B,S,G)
(D,S,G)
(D,T,G)
(B,S,G)



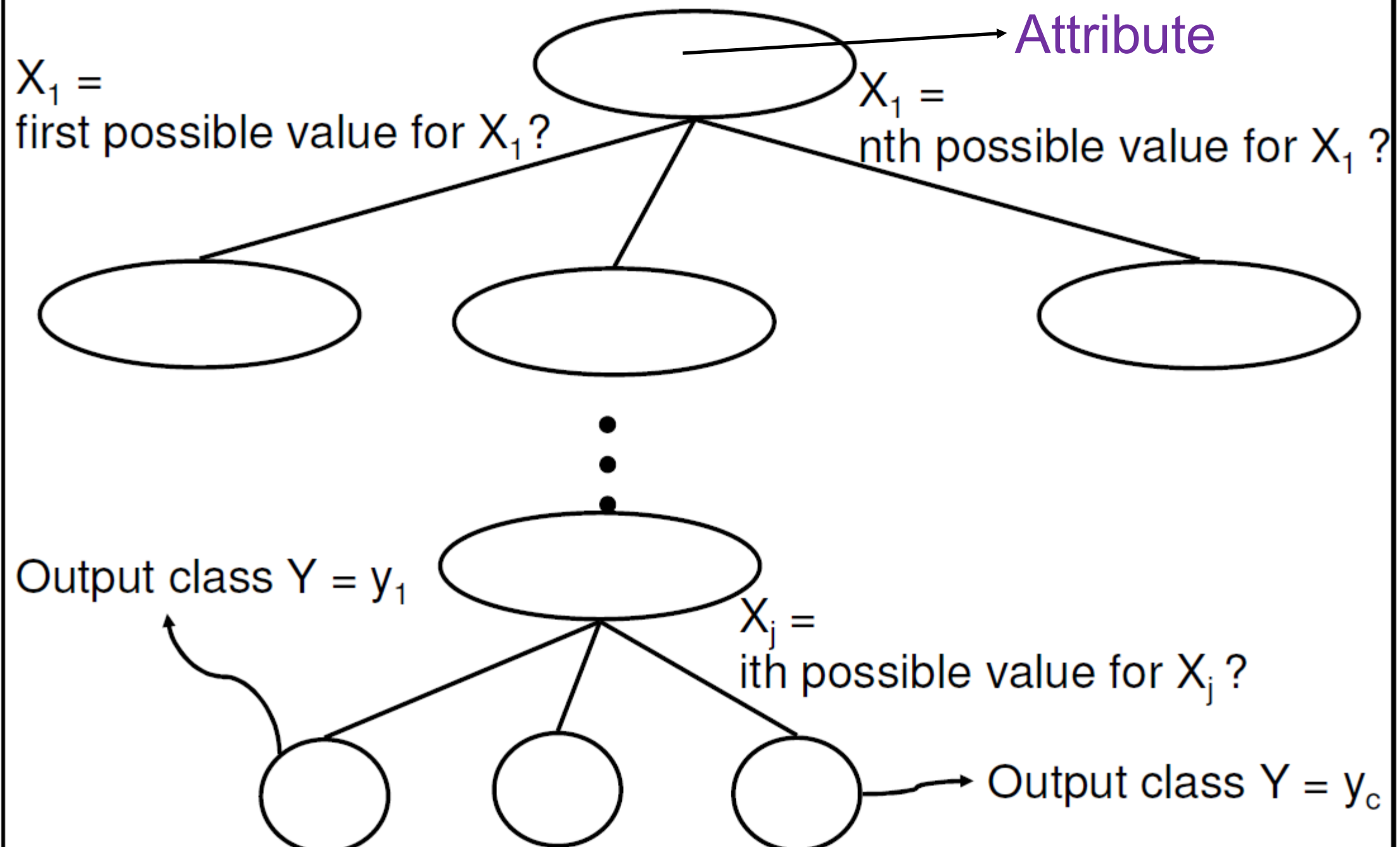
At each level of the tree,
we split the data
according to the value
of one of the attributes

After enough splits, only
one class is represented
in the node → This is a
terminal leaf of the tree
We call that class the
output class for that
node



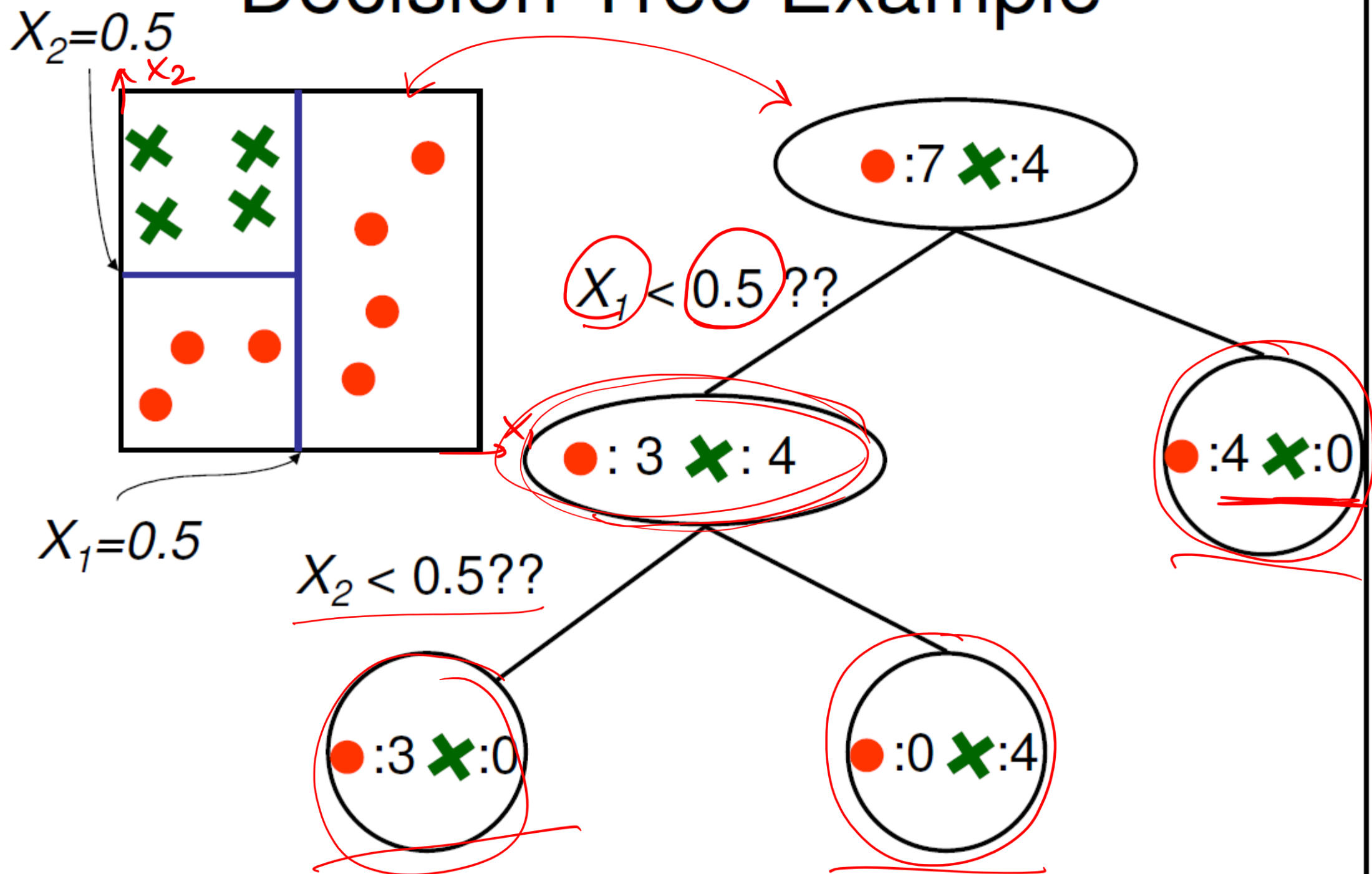
'G' is the output
for this node

General Decision Tree (Discrete Attributes)

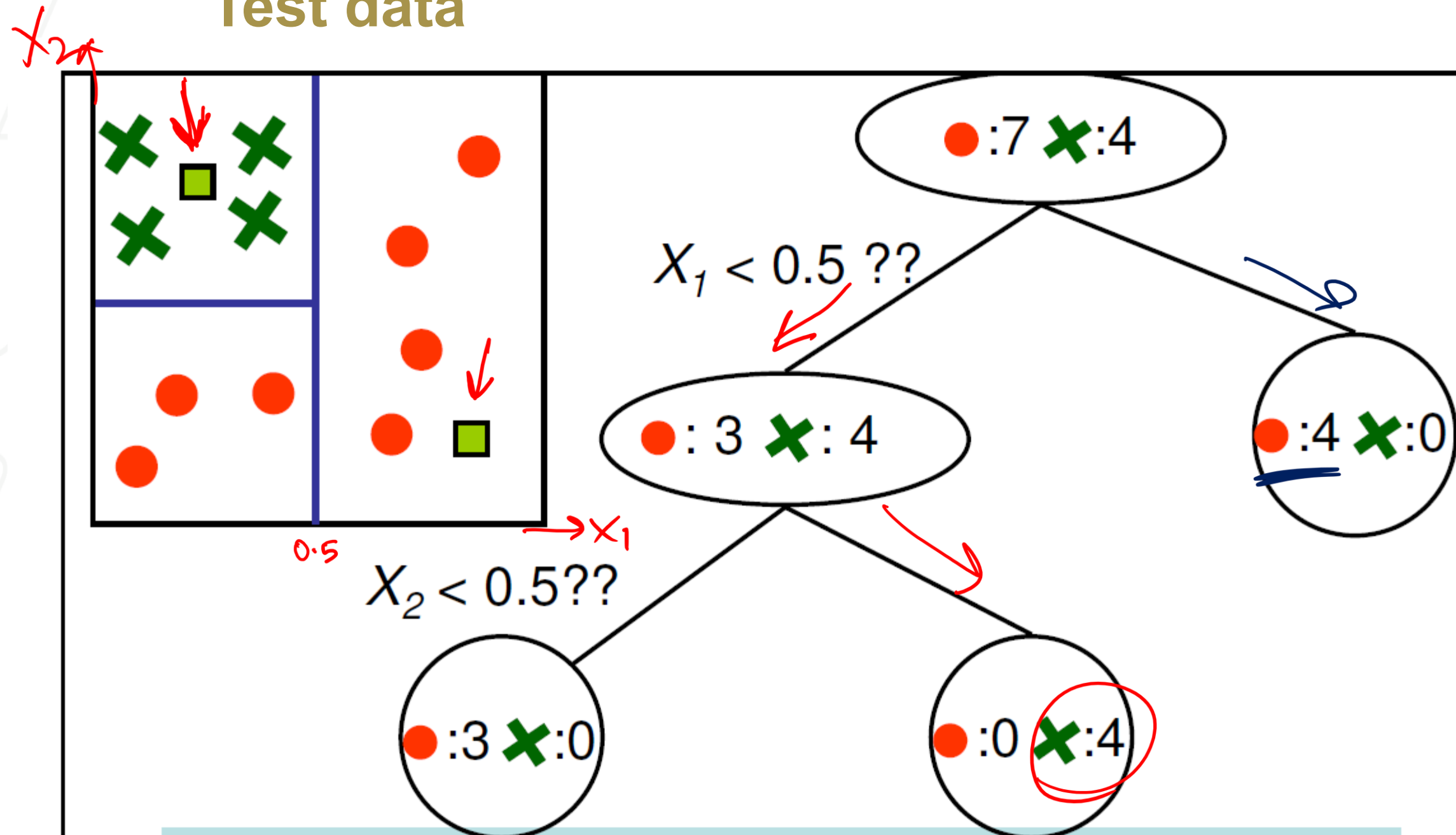


Continuous attributes

Decision Tree Example

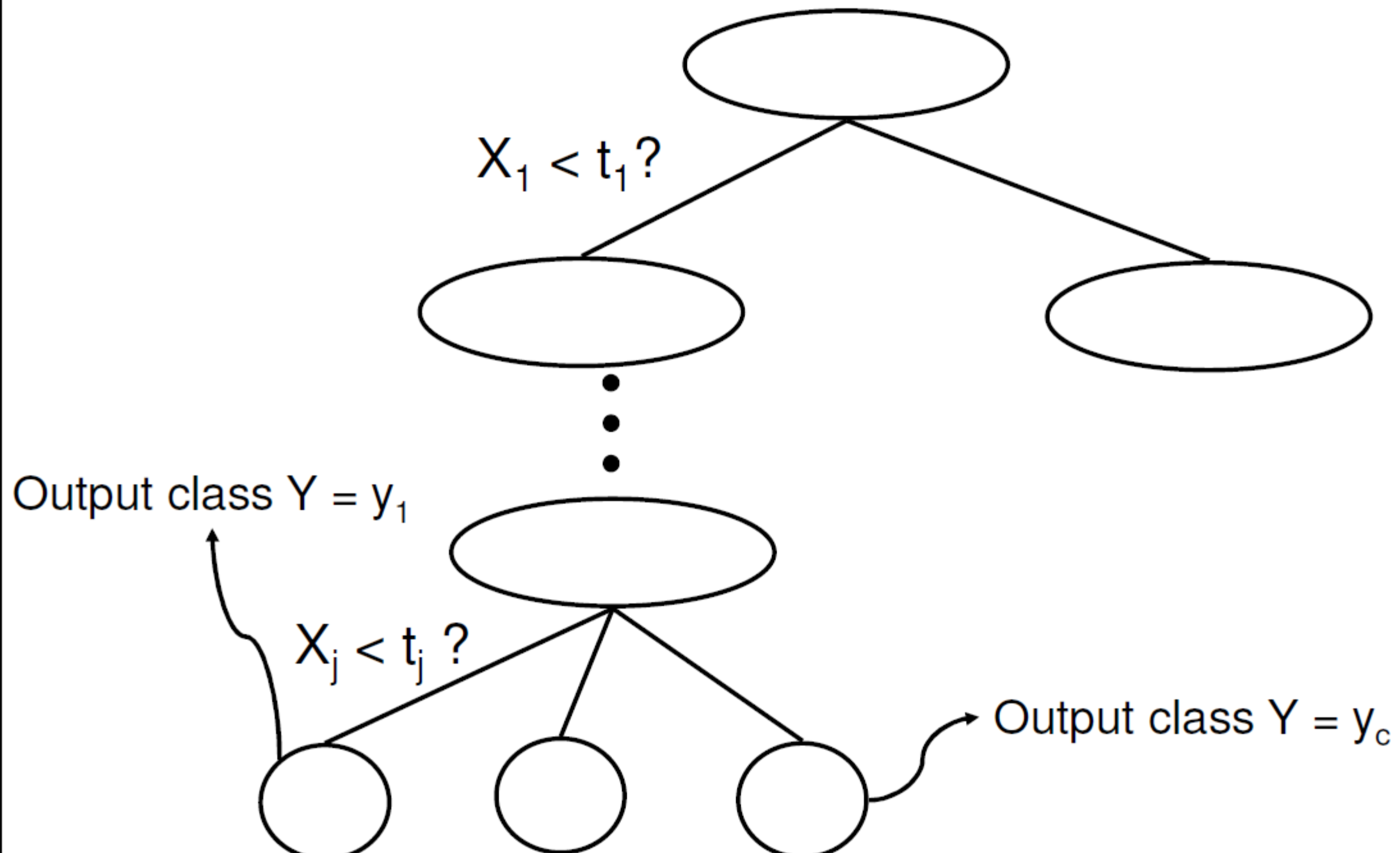


Test data



The class of a new input can be classified by following the tree all the way down to a leaf and by reporting the output of the leaf. For example:
 (0.2,0.8) is classified as $\times$
 (0.8,0.2) is classified as $\bullet$

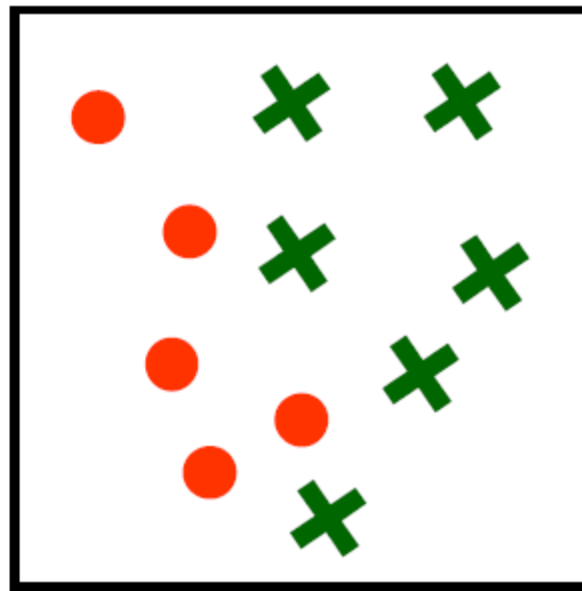
General Decision Tree (Continuous Attributes)



Basic Questions

- How to choose the attribute/value to split on at each level of the tree?
- When to stop splitting? When should a node be declared a leaf?
- If a leaf node is impure, how should the class label be assigned?
- If the tree is too large, how can it be pruned?

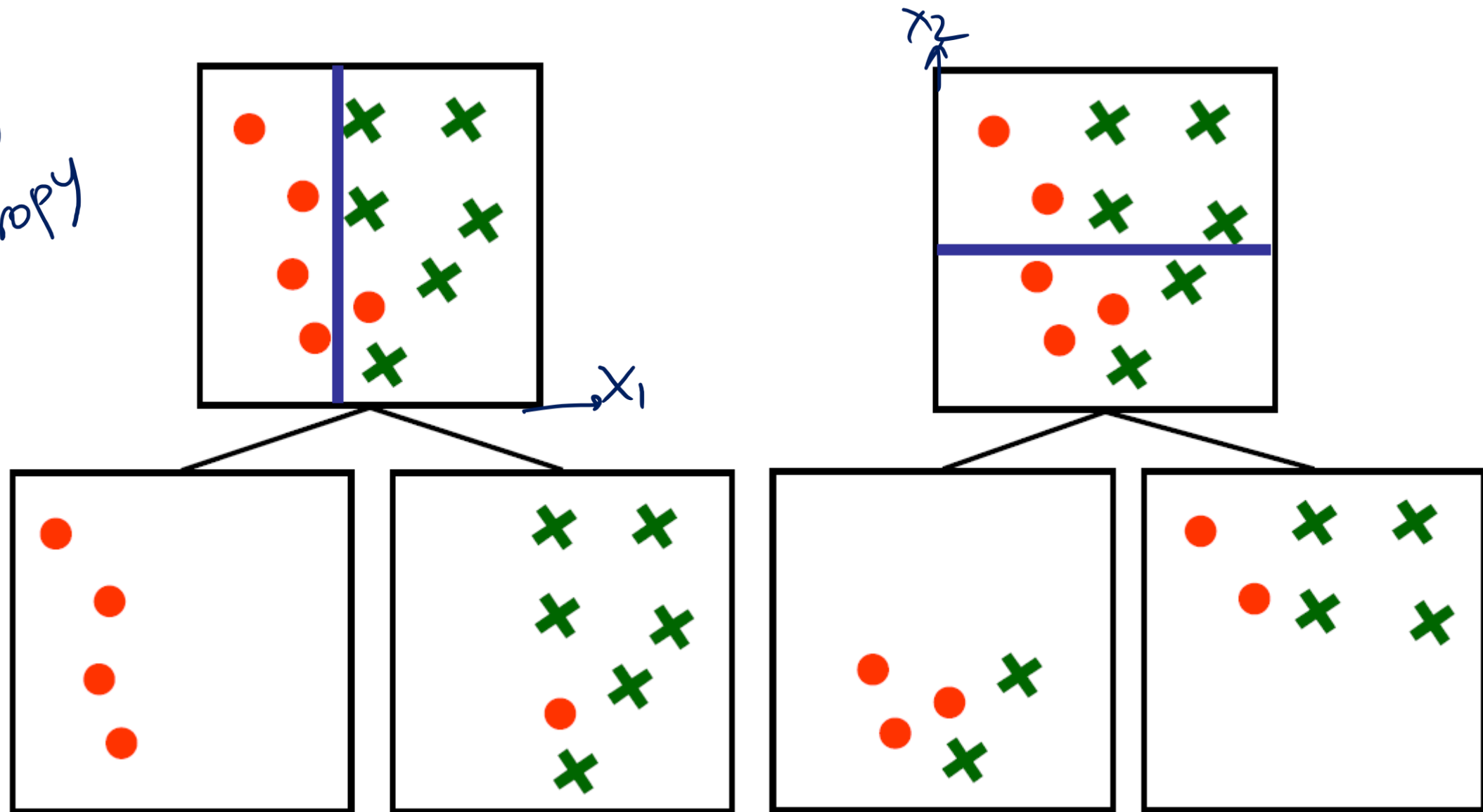
How to choose the attribute/value to split on at each level of the tree?



- Two classes (red circles/green crosses)
- Two attributes: X_1 and X_2
- 11 points in training data
- Idea $\rightarrow$ Construct a decision tree such that the leaf nodes predict correctly the class for all the training examples

How to choose the attribute/value to split on at each level of the tree?

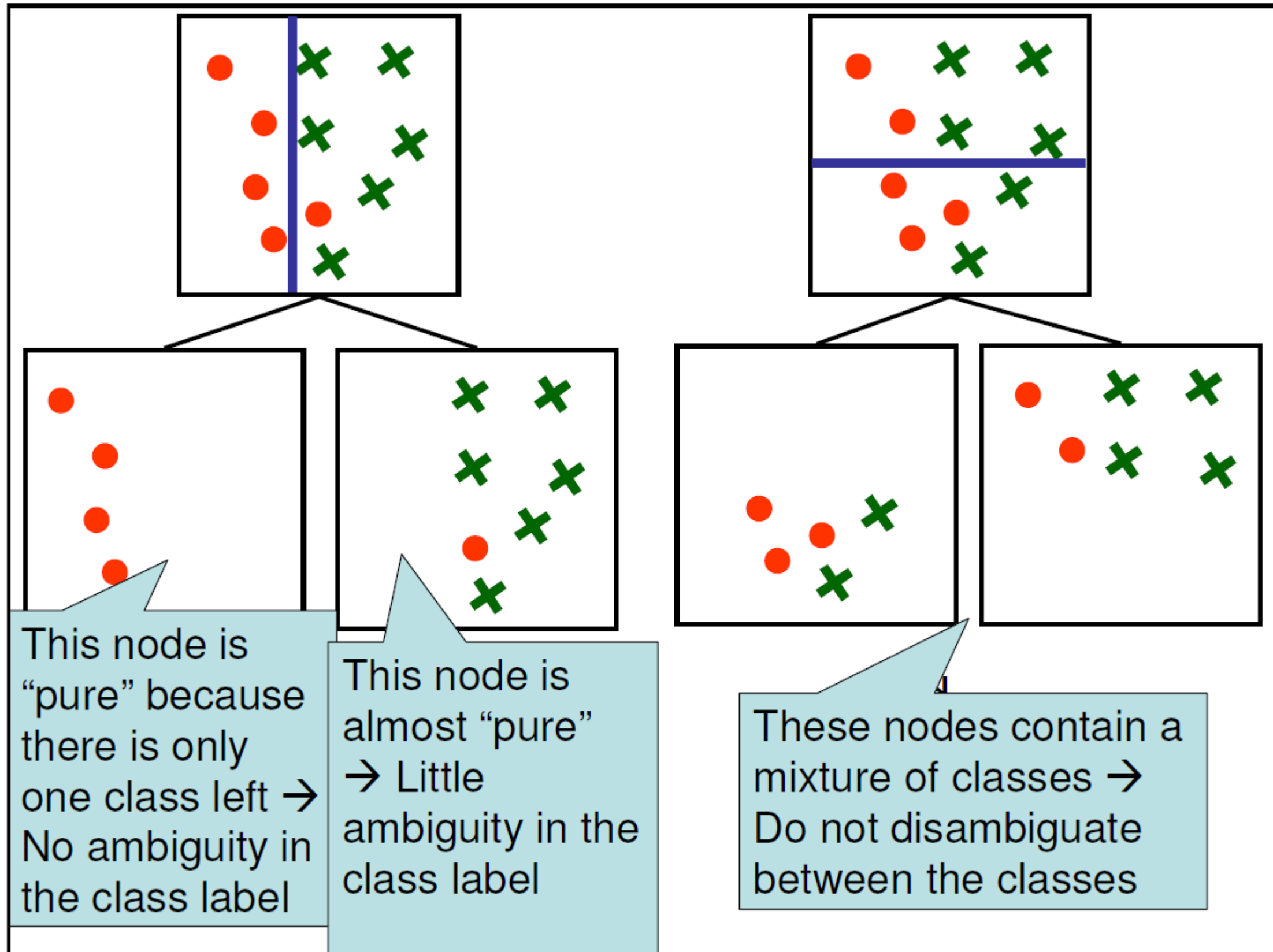
Low Entropy



Good

Bad

We want to find the most compact, smallest size tree (Occam's razor), that classifies the training data correctly → We want to find the split choices that will get us the fastest to pure nodes

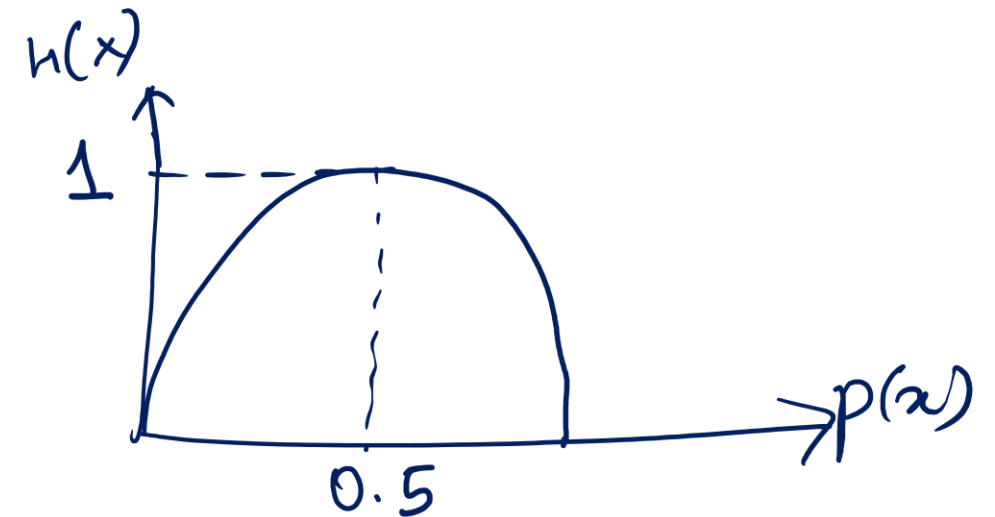


Information Theory Recap

- $I(x) = -\log_2 p(x)$

- $H(x) = E[I(x)] = -\sum p(x) \log_2 p(x)$
Entropy

- Binary Case: Maximum value of entropy is



- Non-binary case: Maximum value of entropy is $\log_2 k$ ← no. of possible states
Fair die $\rightarrow \text{Max } H(x) \rightarrow \log_2 6$

- Range of entropy

$$[0, \infty)$$

Information Content

Coin flip

Increase

C_{1H}	0
C_{1T}	6

$$P(C_{1H}) = 0/6 = 0$$

$$P(C_{1T}) = 6/6 = 1$$

$$\text{Entropy} = 0$$

$$-\sum p(x) \log_2 p(x) = -(0 \log_2 0 + 1 \log_2 1) = 0$$

C_{2H}	1
C_{2T}	5

$$P(C_{2H}) = 1/6$$

$$P(C_{2T}) = 5/6$$

C_{3H}	2
C_{3T}	4

$$P(C_{3H}) = 2/6$$

$$P(C_{3T}) = 4/6$$

Which coin will give us the purest information?

Entropy ~ Uncertainty

Lower uncertainty, higher information gain

$$H(X) = - \sum_{i=1}^N P(x=i) \log_2 P(x=i)$$

$$\text{Entropy} = -0 \log 0 - 1 \log 1 = -0 - 0 = 0$$

$$\text{Entropy} = - (1/6) \log_2 (1/6) - (5/6) \log_2 (5/6) = 0.65$$

$$\text{Entropy} = - (2/6) \log_2 (2/6) - (4/6) \log_2 (4/6) = 0.92$$

Entropy

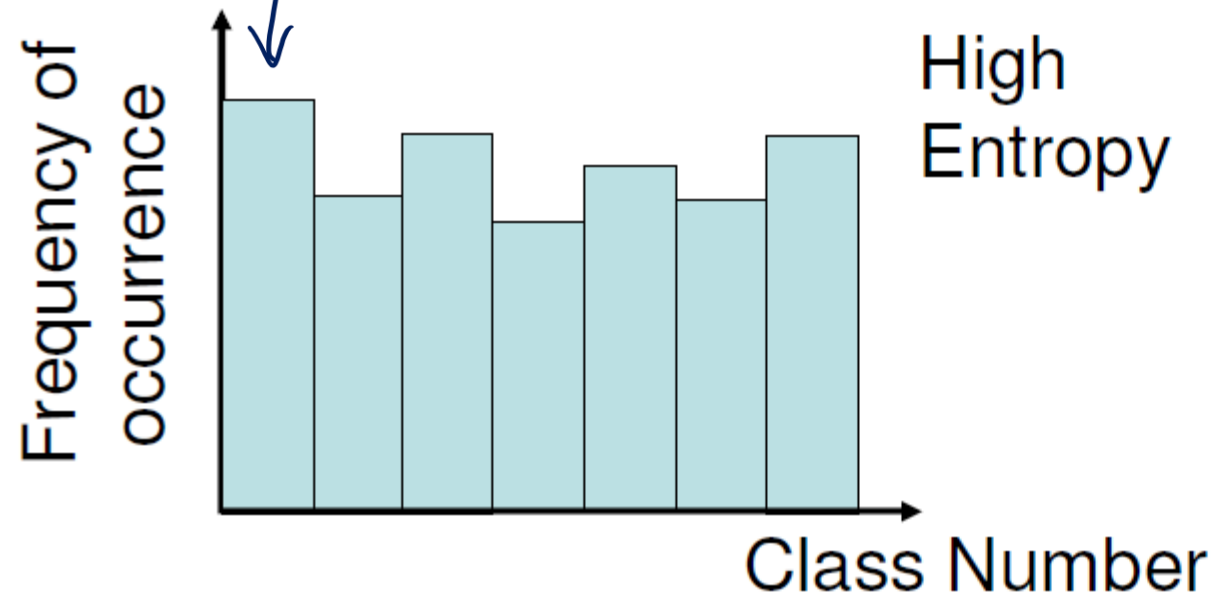
- In general, the average number of bits necessary to encode n values is the entropy:

$$H = - \sum_{i=1}^n P_i \log_2 P_i$$

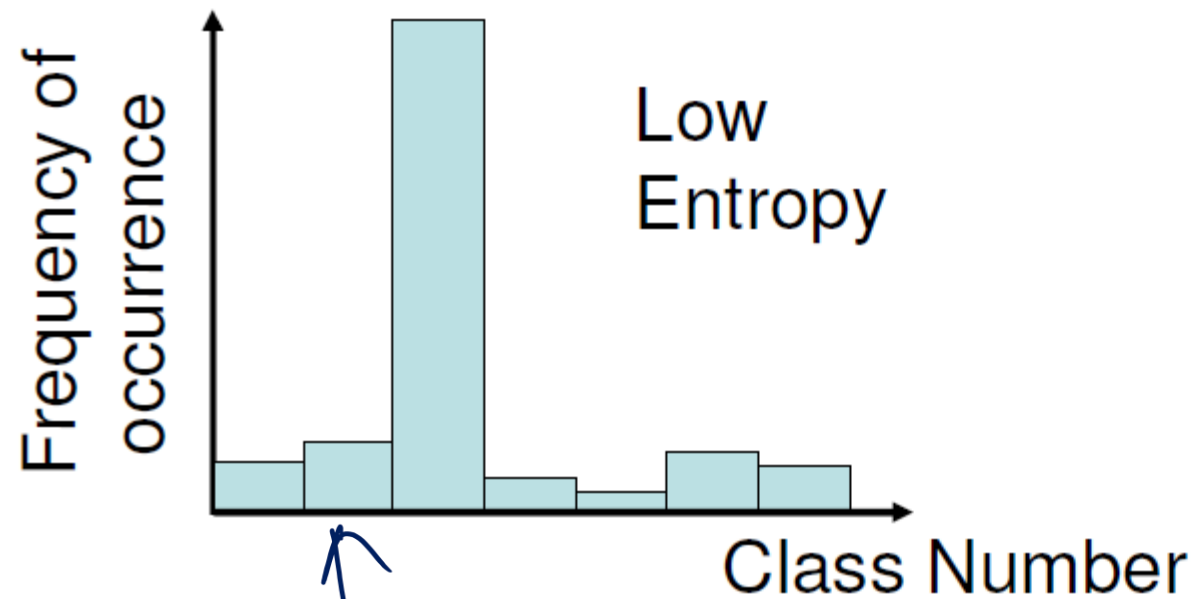
- P_i = probability of occurrence of value i
 - High entropy $\rightarrow$ All the classes are (nearly) equally likely
 - Low entropy $\rightarrow$ A few classes are likely; most of the classes are rarely observed

Uniform (Higher Entropy)

Entropy



The entropy captures the degree of “purity” of the distribution



Normal

$$-\sum p(x) \log_2 p(x) = -\left(\frac{1}{7} \log \frac{1}{7} + \frac{6}{7} \log \frac{6}{7}\right)$$

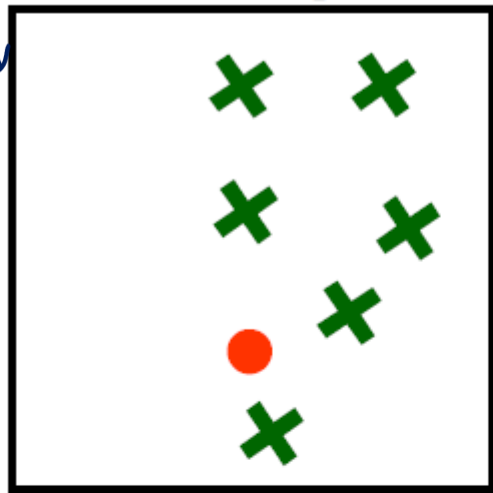
$p_{\text{red}} = 1/7 \quad p_{\text{green}} = 6/7$

$$-\left(\frac{2}{5} \log \frac{2}{5} + \frac{3}{5} \log \frac{3}{5}\right)$$

$$-\left(\frac{1}{4} \log \frac{1}{4} + \frac{3}{4} \log \frac{3}{4}\right)$$

Example Entropy Calculation

①



$$N_A = 1$$

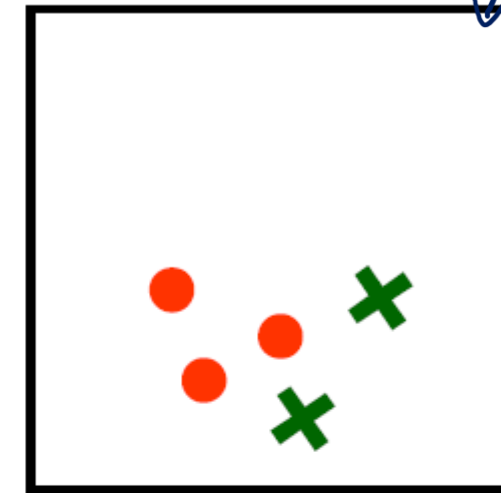
$$N_B = 6$$

$$p_A = N_A / (N_A + N_B) = 1/7$$

$$p_B = N_B / (N_A + N_B) = 6/7$$

$$H_1 = -p_A \log_2 p_A - p_B \log_2 p_B = 0.59$$

②



$$N_A = 3$$

$$N_B = 2$$

$$p_A = N_A / (N_A + N_B) = 3/5$$

$$p_B = N_B / (N_A + N_B) = 2/5$$

$$H_2 = -p_A \log_2 p_A - p_B \log_2 p_B = 0.97$$

$$H_1 < H_2 \Rightarrow (2) \text{ less pure than } (1)$$

Cats & dogs
50 50

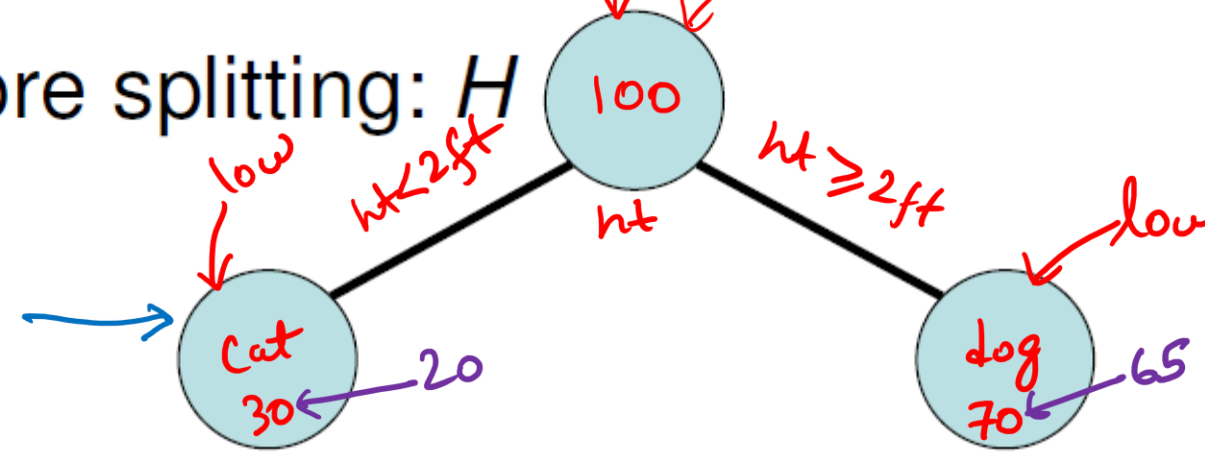
Entropy high
1

Average entropy
after split

Conditional Entropy

$$= E[H(x)]$$
$$= \sum p(x) H(x)$$

Entropy before splitting: H



After splitting, a fraction P_L of the data goes to the left node, which has entropy H_L

After splitting, a fraction P_R of the data goes to the right node, which has entropy H_R

The average entropy after splitting is:

Entropy of left
node

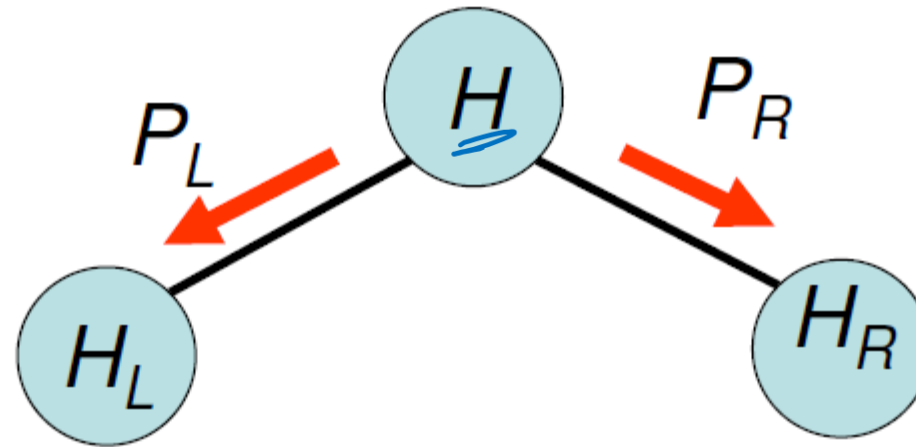
$$H_L \times P_L + H_R \times P_R$$

“Conditional Entropy”

Probability that a random input
is directed to the left node

rgia
h.

Information Gain



We want nodes as pure as possible

→ We want to reduce the entropy as much as possible

→ We want to maximize the difference between the entropy of the parent node and the expected entropy of the children

Information Gain (IG) = Amount by which the ambiguity is decreased by splitting the node

Maximize:

$$IG = H - (H_L \times P_L + H_R \times P_R)$$

Maximize

Information Gain

average entropy after splitting

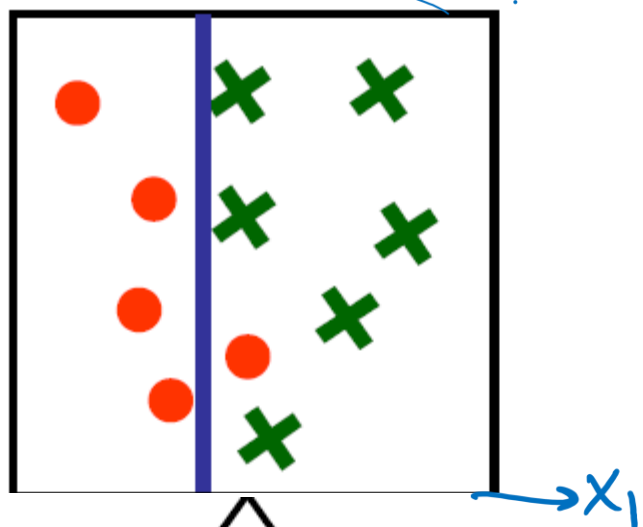
Notations

- Entropy: $H(Y)$ = Entropy of the distribution of classes at a node
- Conditional Entropy:
 - *Discrete*: $H(Y|X_j)$ = Entropy after splitting with respect to variable j
 - *Continuous*: $H(Y|X_j, t)$ = Entropy after splitting with respect to variable j with threshold t
- Information gain:
 - *Discrete*: $IG(Y|X_j) = H(Y) - H(Y|X_j)$ = Entropy after splitting with respect to variable j
 - *Continuous*: $IG(Y|X_j, t) = H(Y) - H(Y|X_j, t)$ = Entropy after splitting with respect to variable j with threshold t

Before
Split:

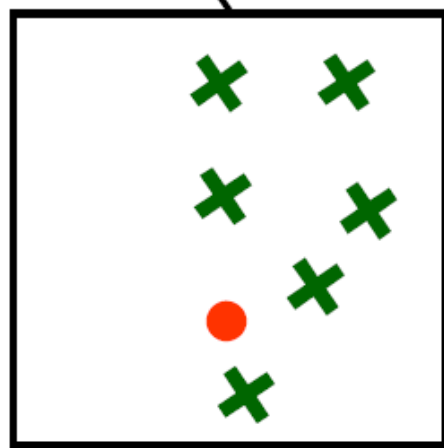
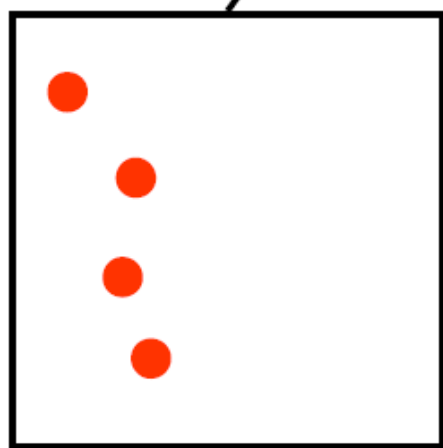
$$P_r = \frac{5}{11} \quad P_g = \frac{6}{11} \quad H(bs) = - \left(\frac{5}{11} \log_2 \frac{5}{11} + \frac{6}{11} \log_2 \frac{6}{11} \right)$$

$$H = 0.99$$



$P_L =$
 $P_R =$
 $H_L =$
 $H_R =$

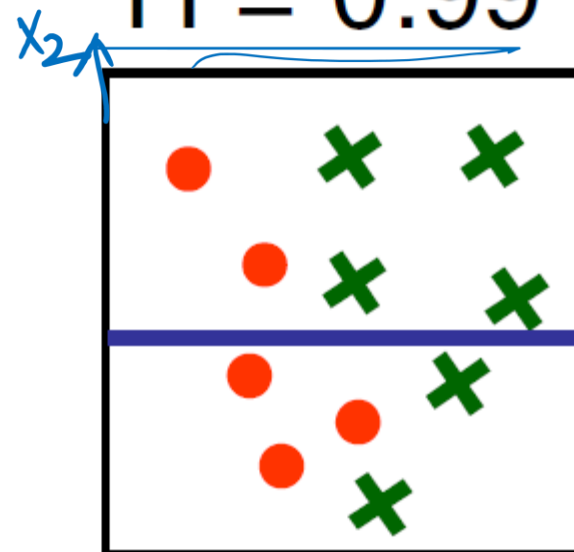
$$IG = H - (H_L * 4/11 + H_R * 7/11)$$



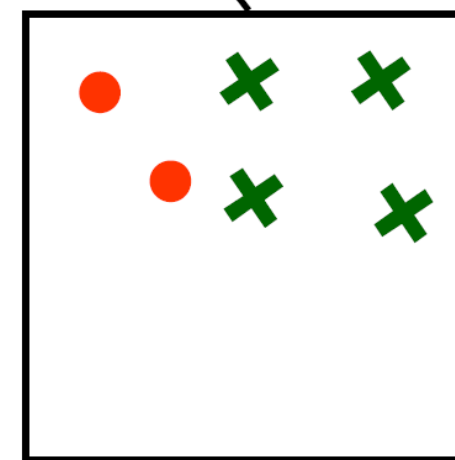
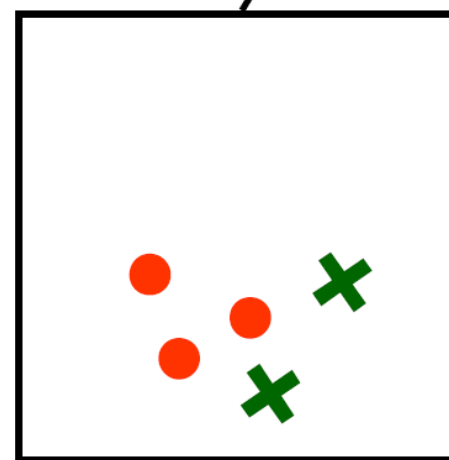
$$H_L = 0$$

$$H_R = 0.58$$

$$H = 0.99$$



$$IG = H - (H_L * 5/11 + H_R * 6/11)$$

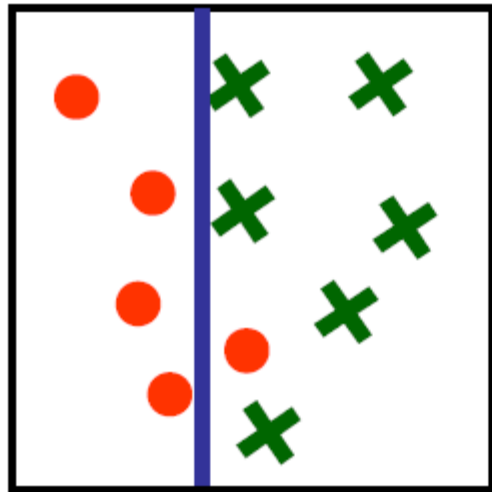


$$H_L = 0.97$$

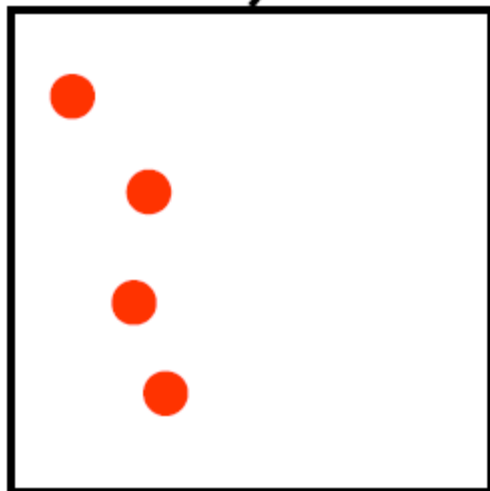
$$H_R = 0.92$$

$\text{np.linspace}(1, 10, 100)$
 f_{low} f_{high}

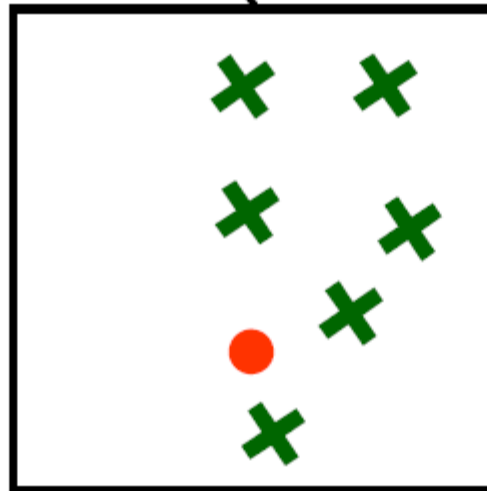
$H = 0.99$



$IG = 0.62$

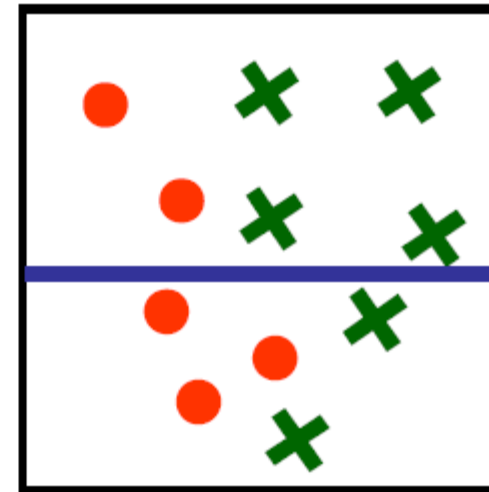


$H_L = 0$

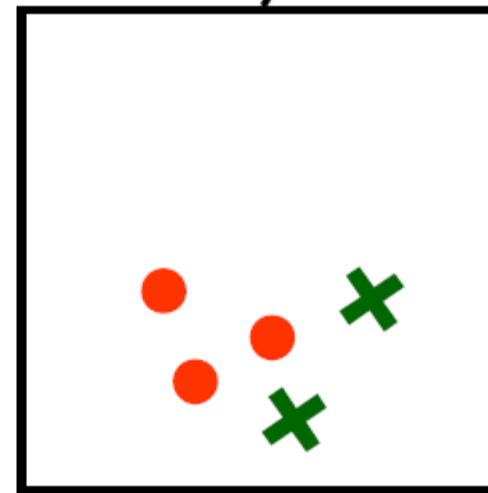


$H_R = 0.58$

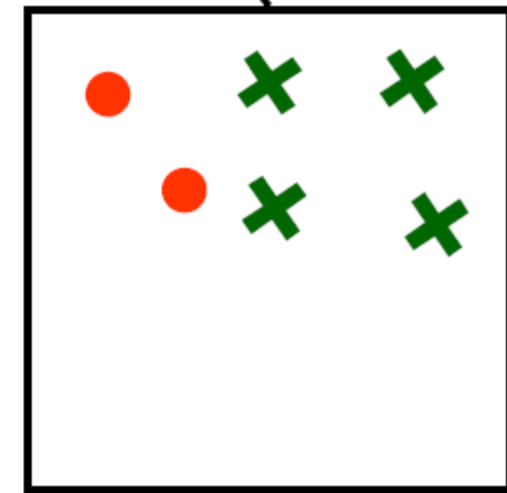
$H = 0.99$



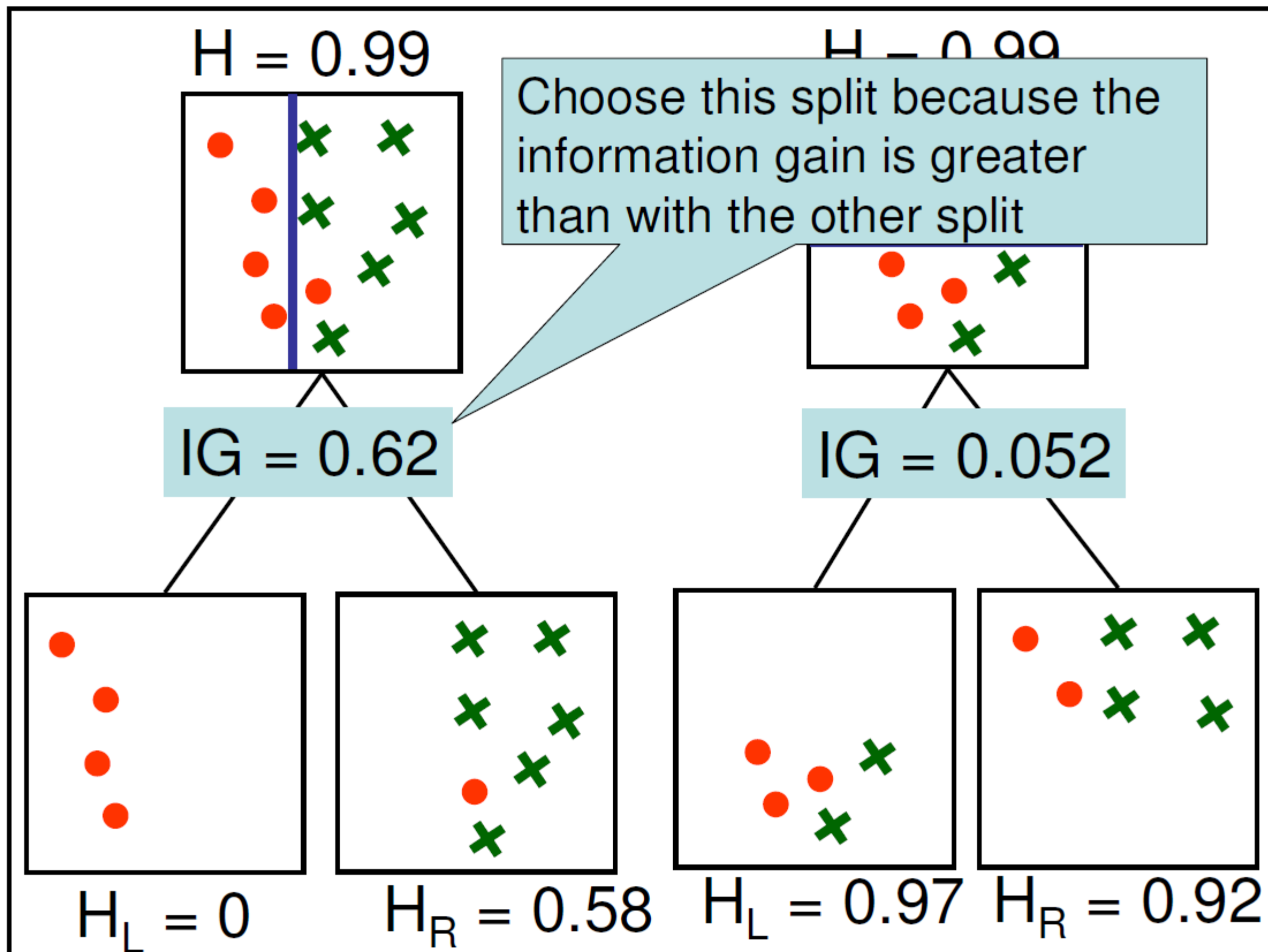
$IG = 0.052$



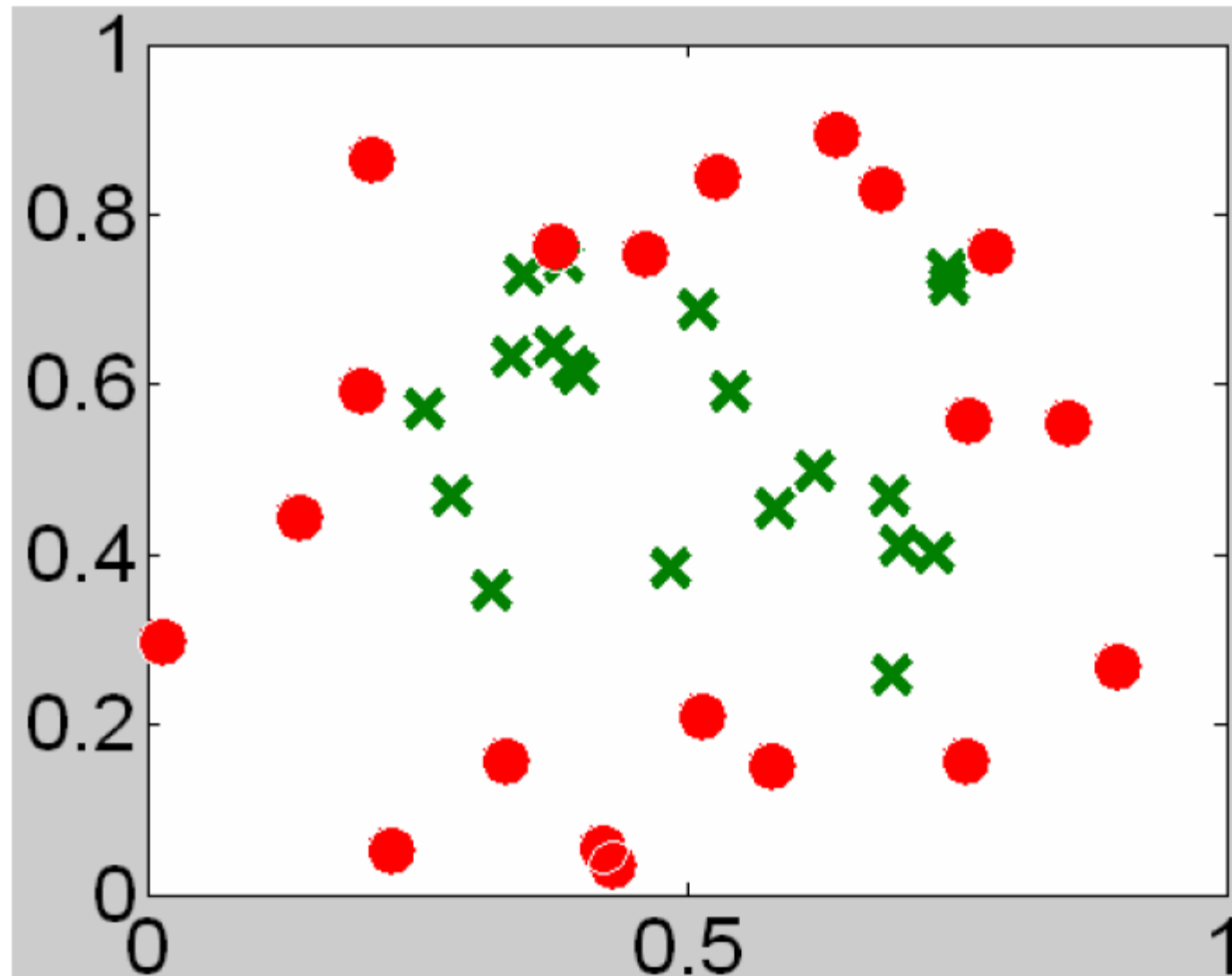
$H_L = 0.97$



$H_R = 0.92$



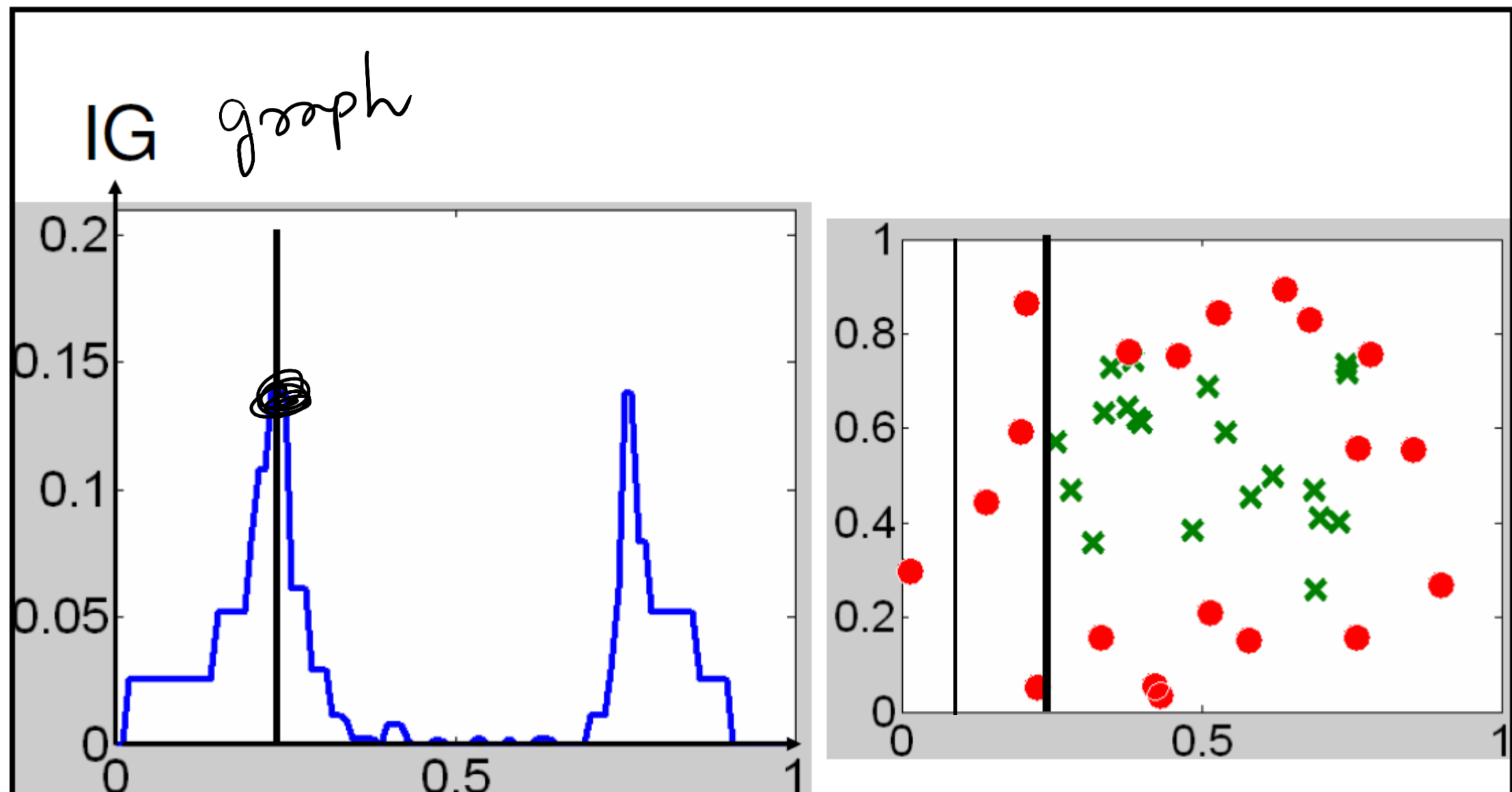
More Complete Example



● = 20 training examples from class A

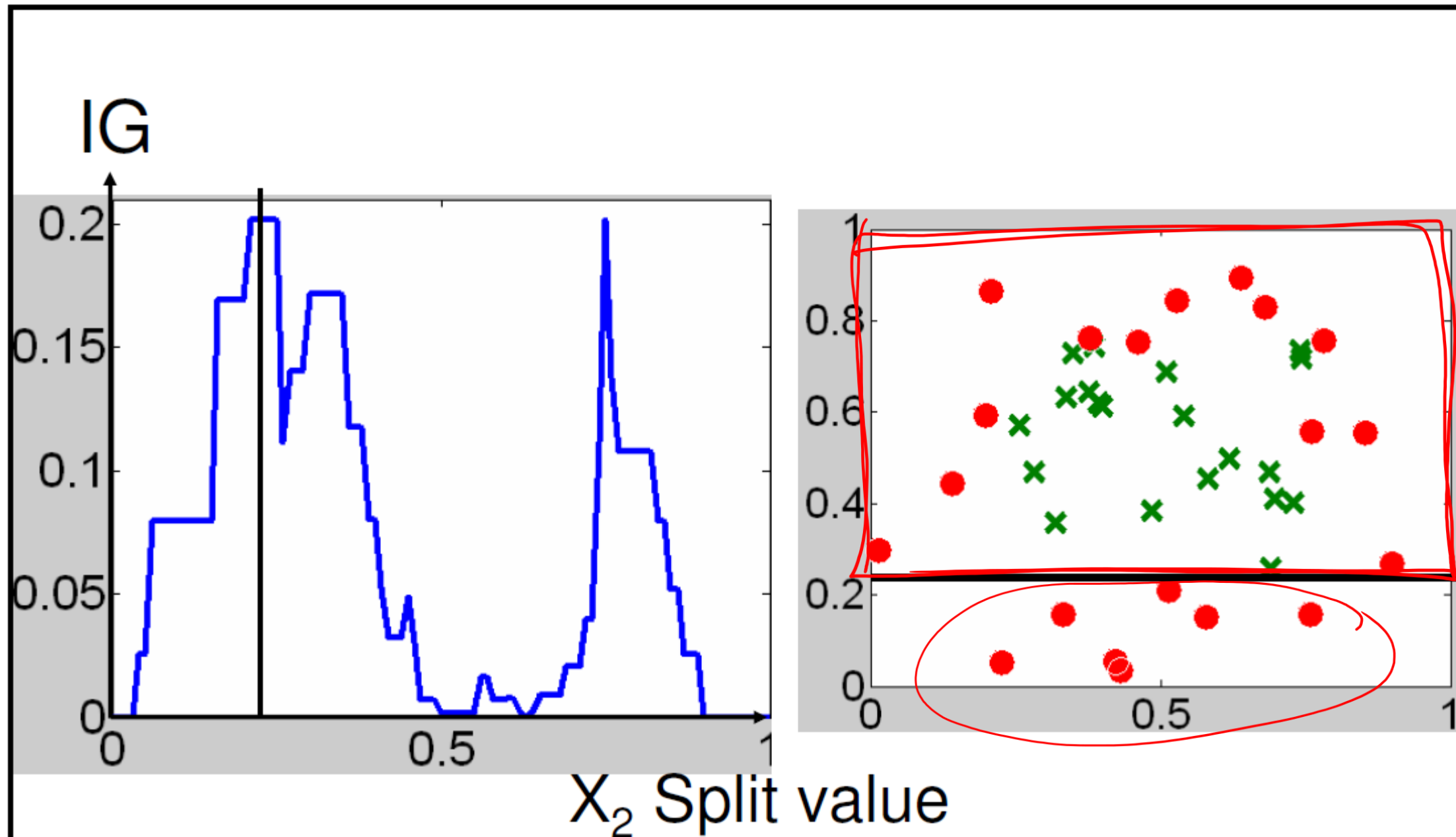
× = 20 training examples from class B

Attributes = X_1 and X_2 coordinates



X_1 Split value

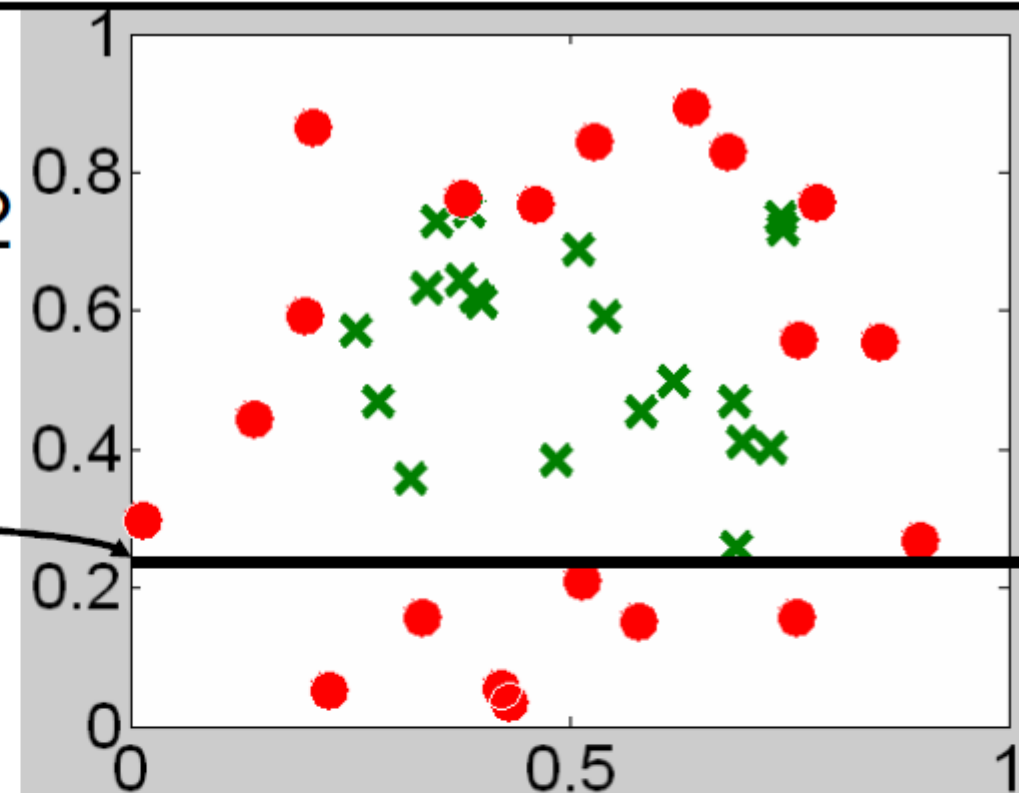
Best split value (max Information Gain) for X_1
attribute: 0.24 with IG = 0.138



Best split value (max Information Gain) for X_2 attribute: 0.234 with $IG = 0.202$

Best X_1 split: 0.24, IG = 0.138
Best X_2 split: 0.234, IG = 0.202

Split on X_2 with 0.234



$X_2 \leq 0.233657$

Total data points = 7
7 A
0 B

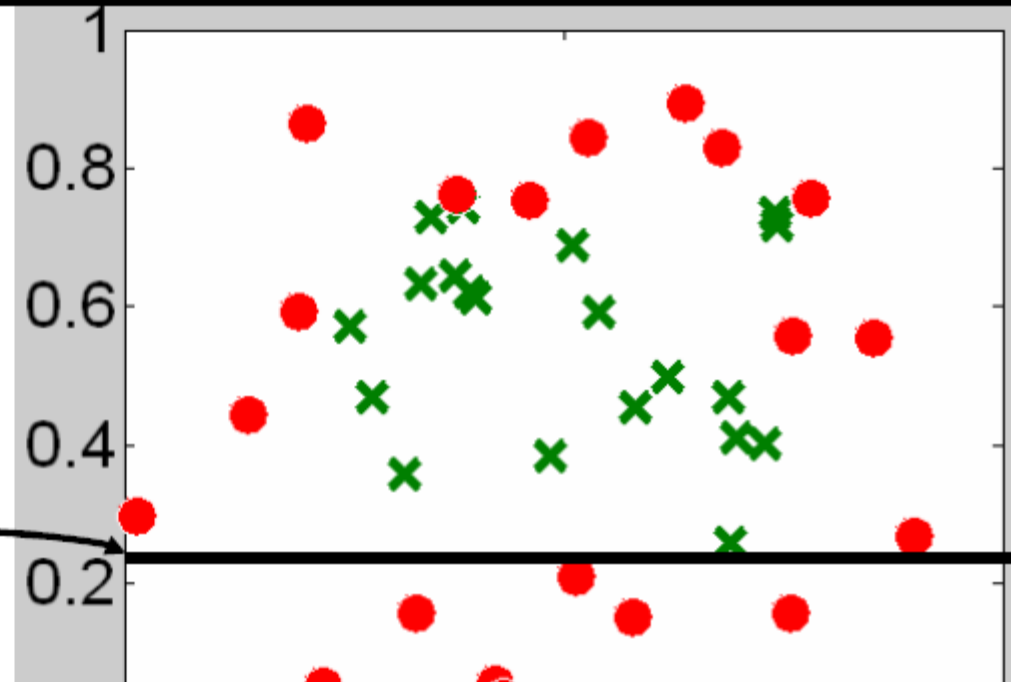
Total data points = 33
13 A
20 B

Left direction for smaller value, right direction for bigger value

Best X split: 0.24 IG = 0.138

Best split: 0.202

There is no point in splitting this node further since it contains only data from a single class → return it as a leaf node with output 'A'



This node is not pure so we need to split further

$X_2 \leq 0.233$

Total data points = 7

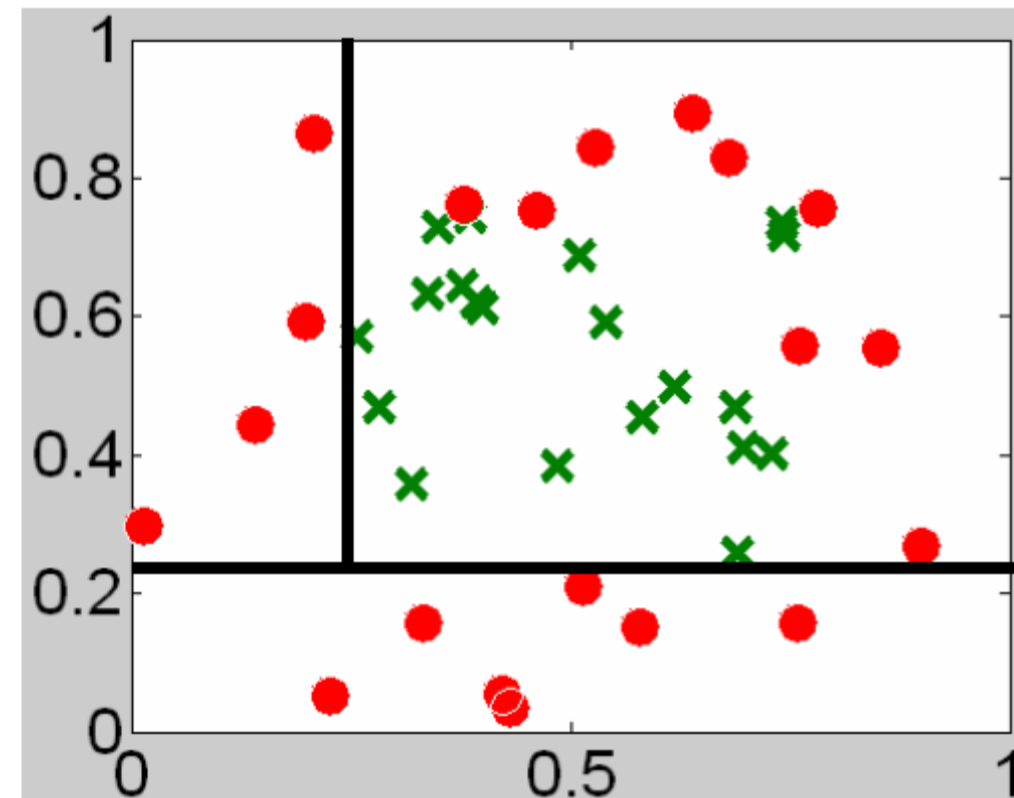
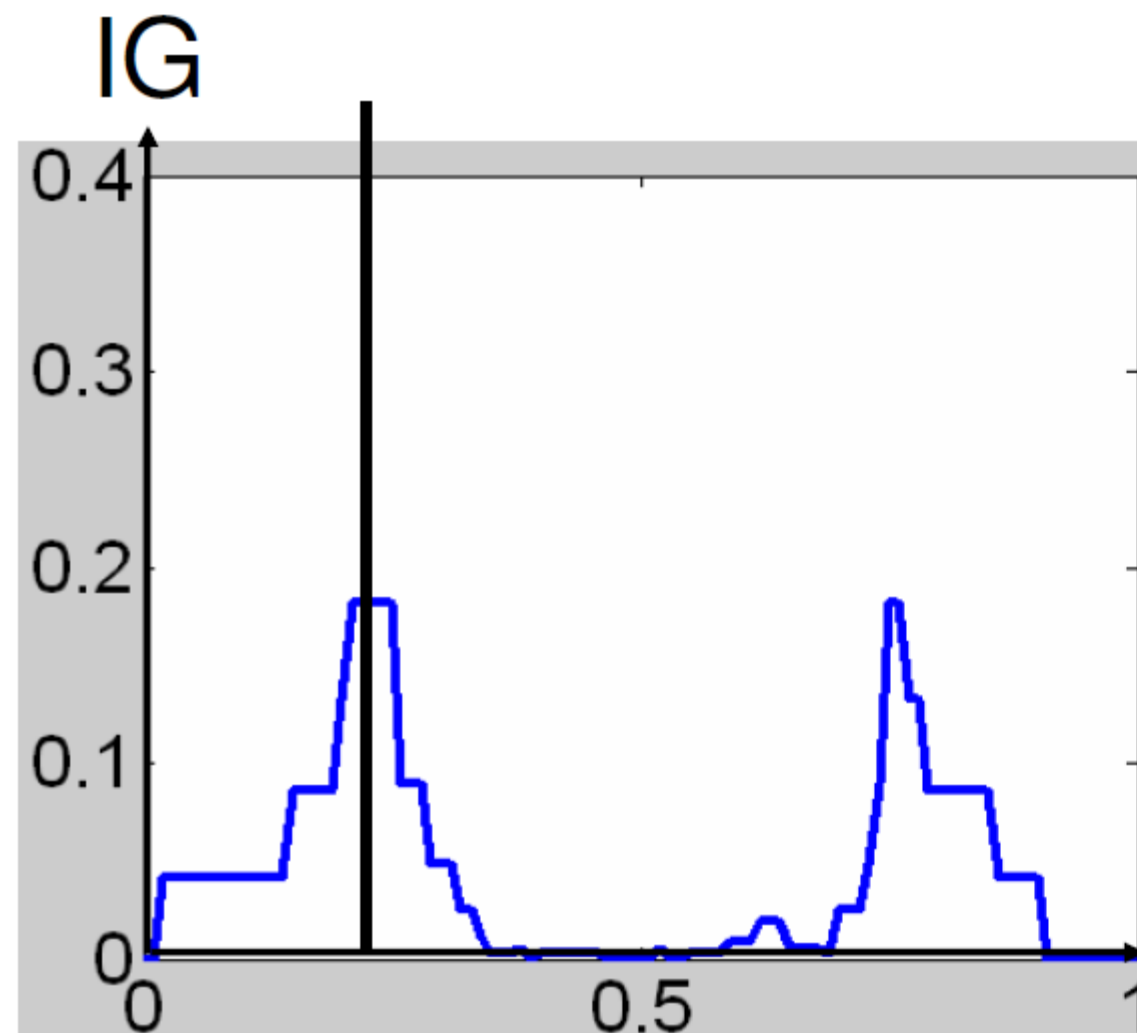
7 A

0 B

Total data points = 33

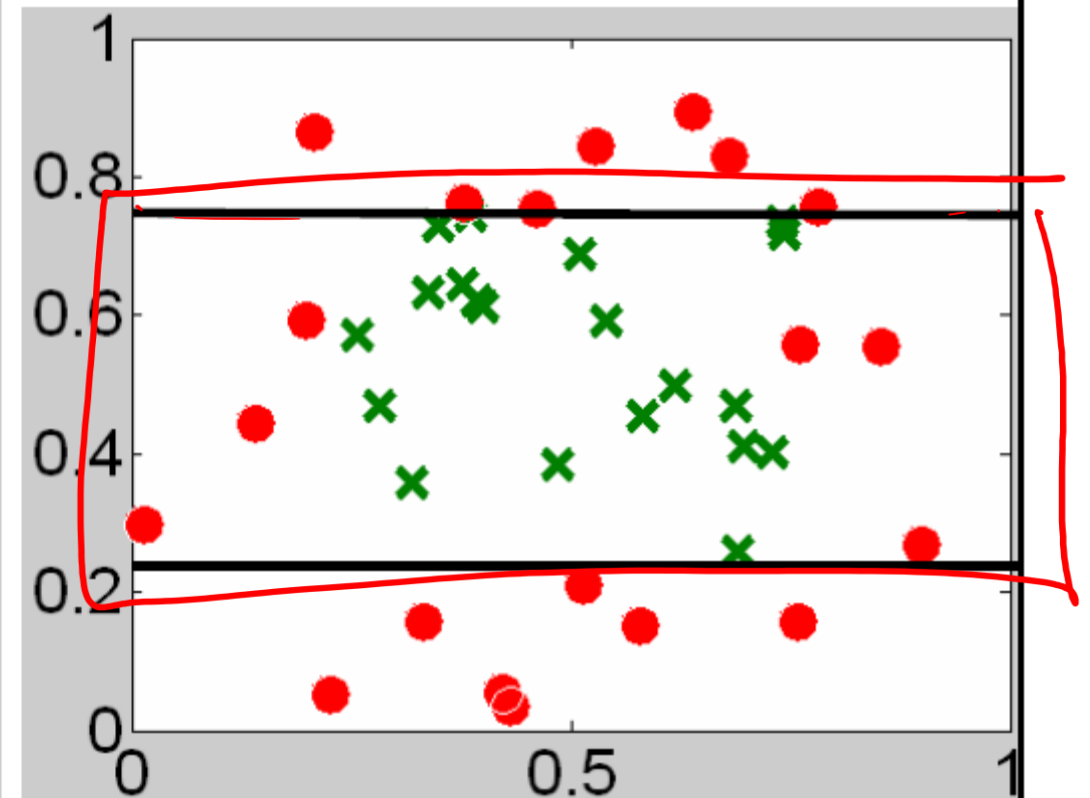
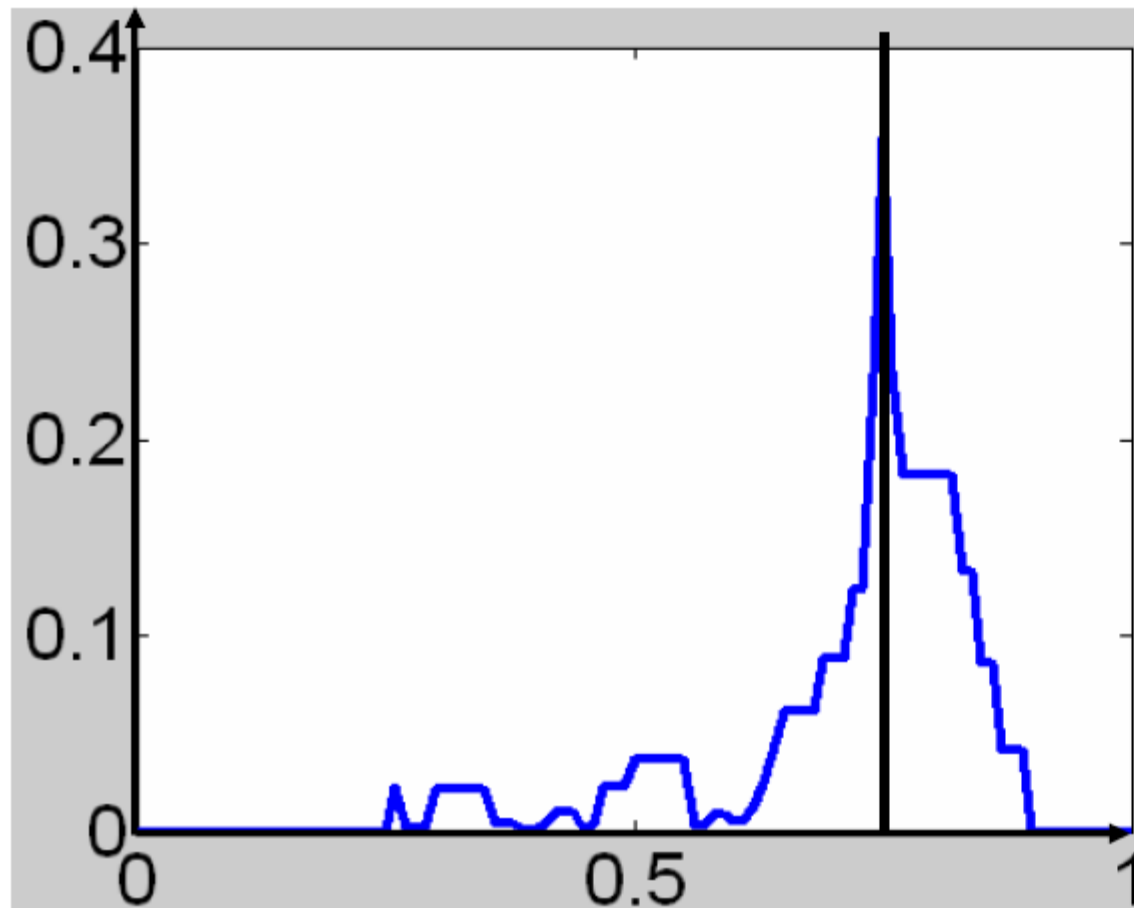
13 A

20 B



Best split value (max Information Gain) for X_1 attribute: 0.22 with IG ~ 0.182

IG

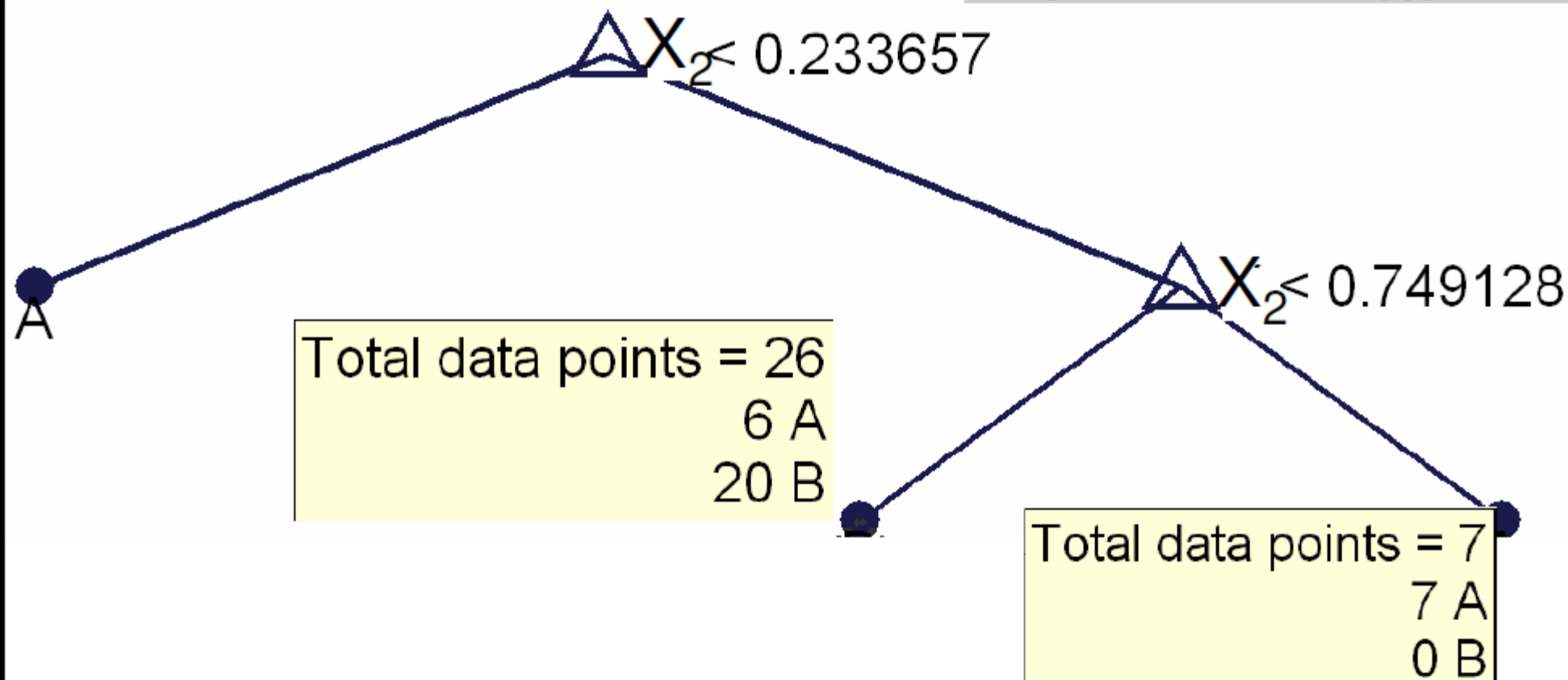
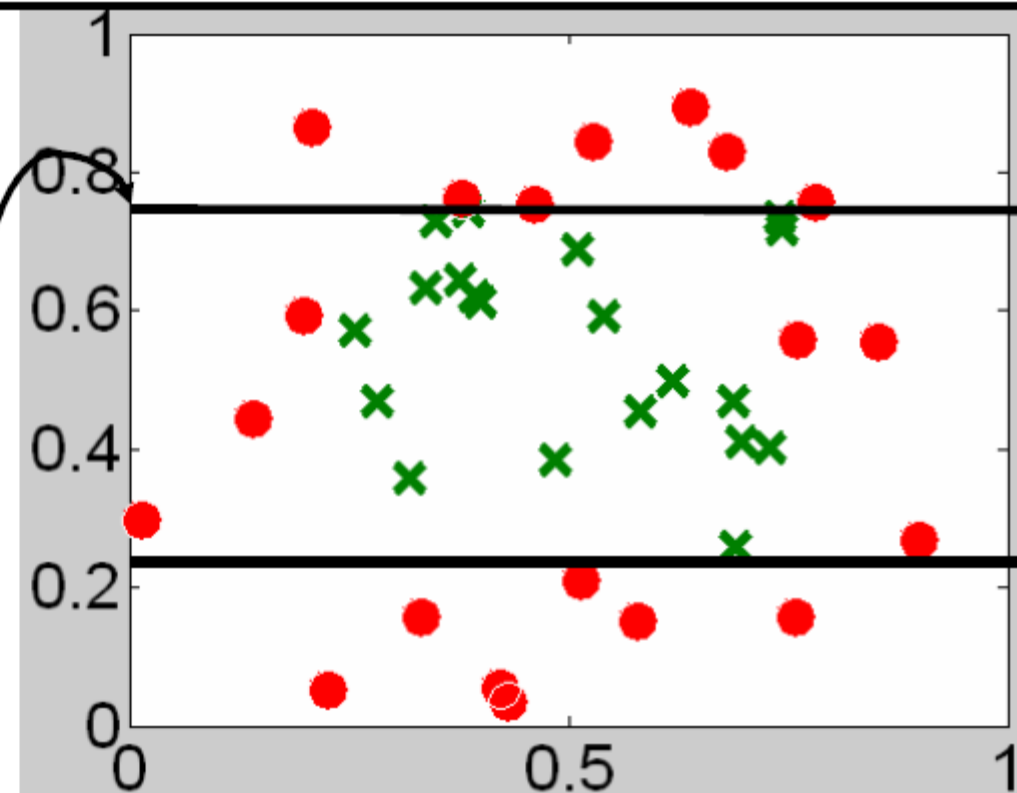


X_2 Split value

Best split value (max Information Gain) for X_2 attribute: 0.75 with IG $\sim$ 0.353

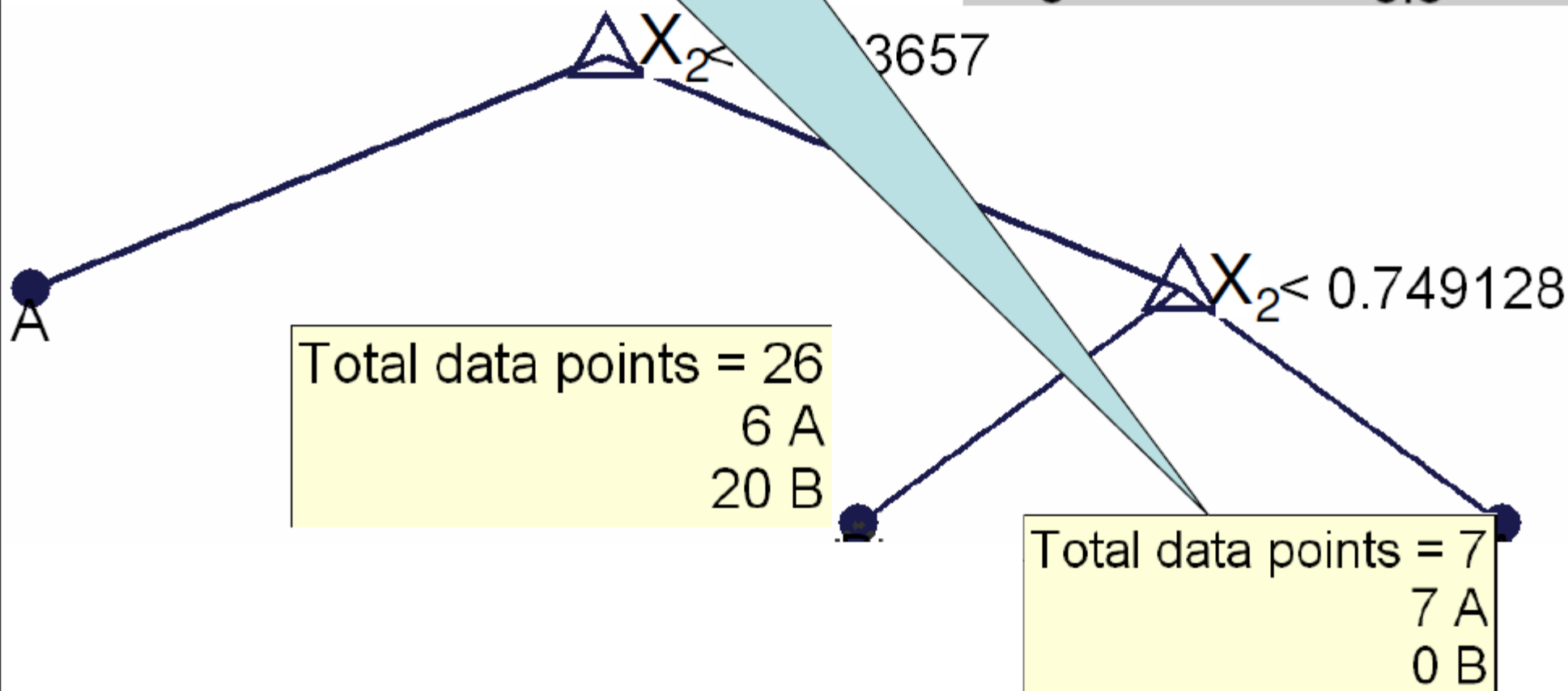
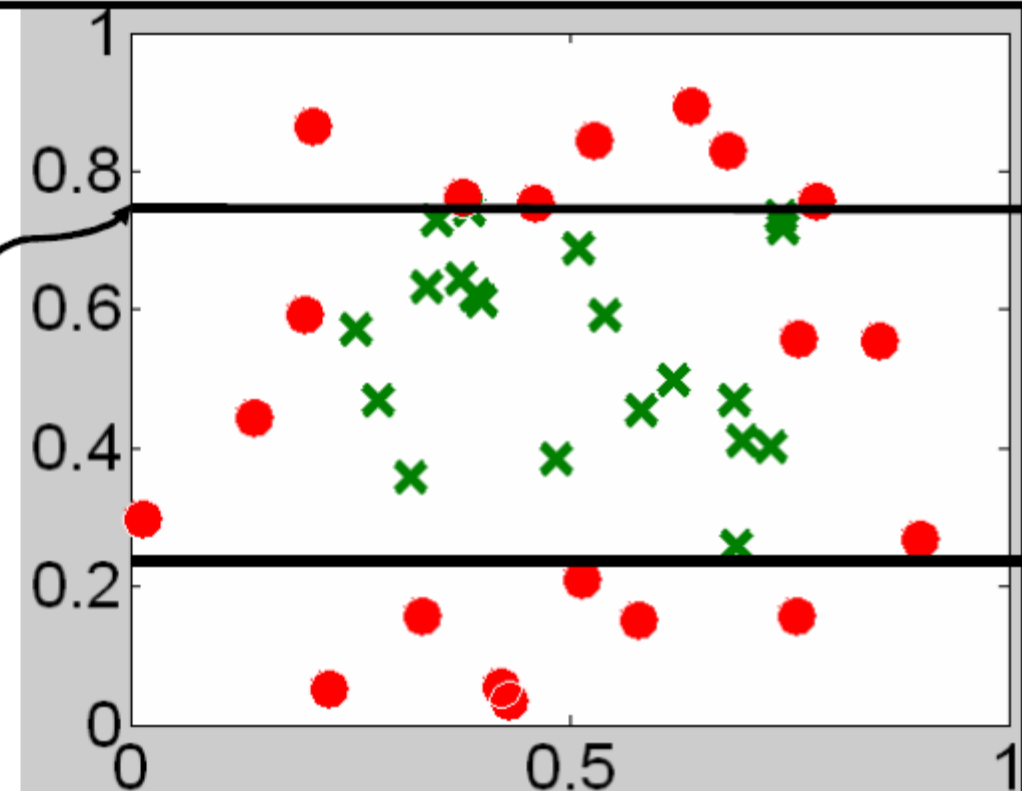
Best X_1 split: 0.22, IG = 0.182
Best X_2 split: 0.75, IG = 0.353

Split on X_2 with 0.75

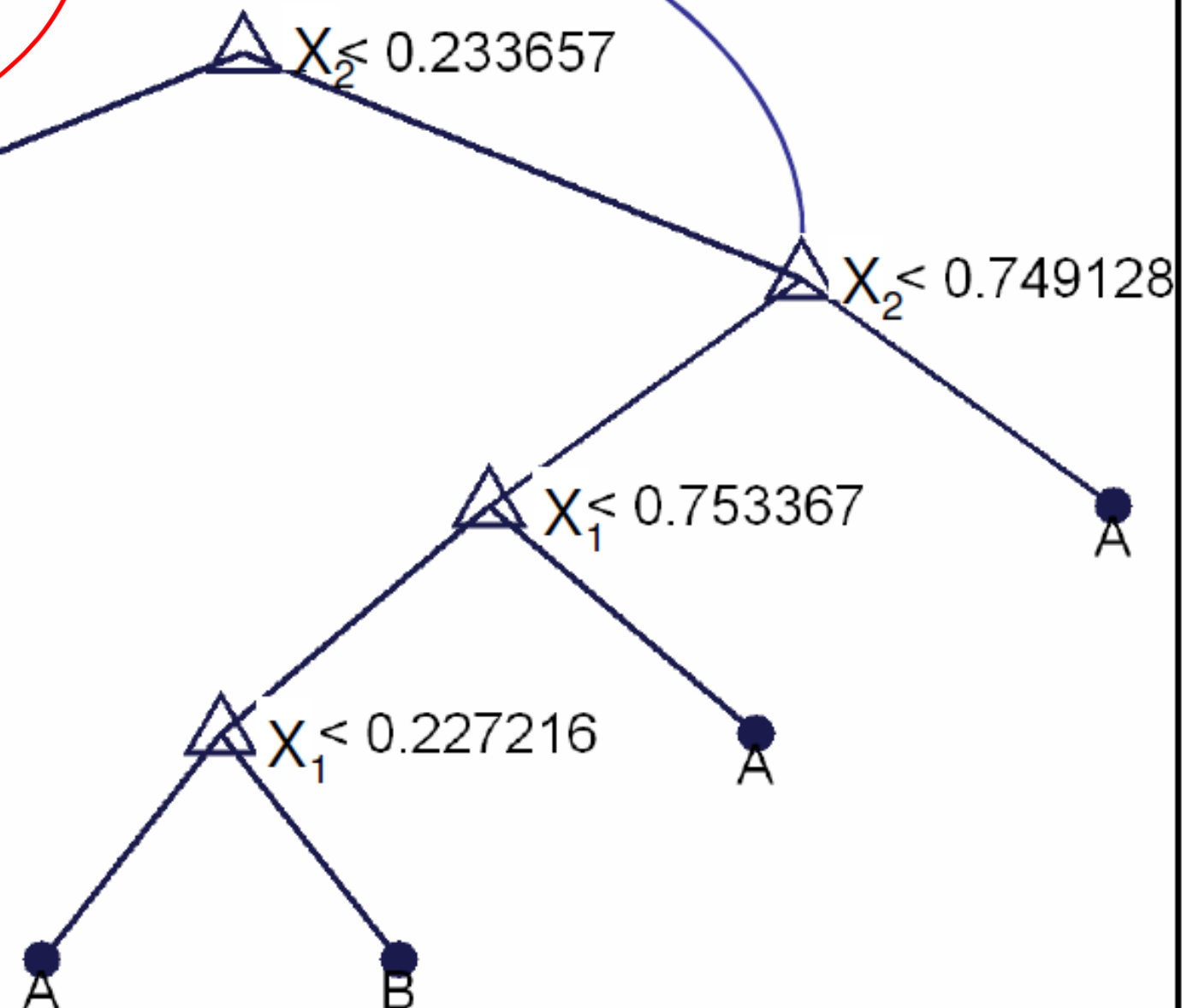
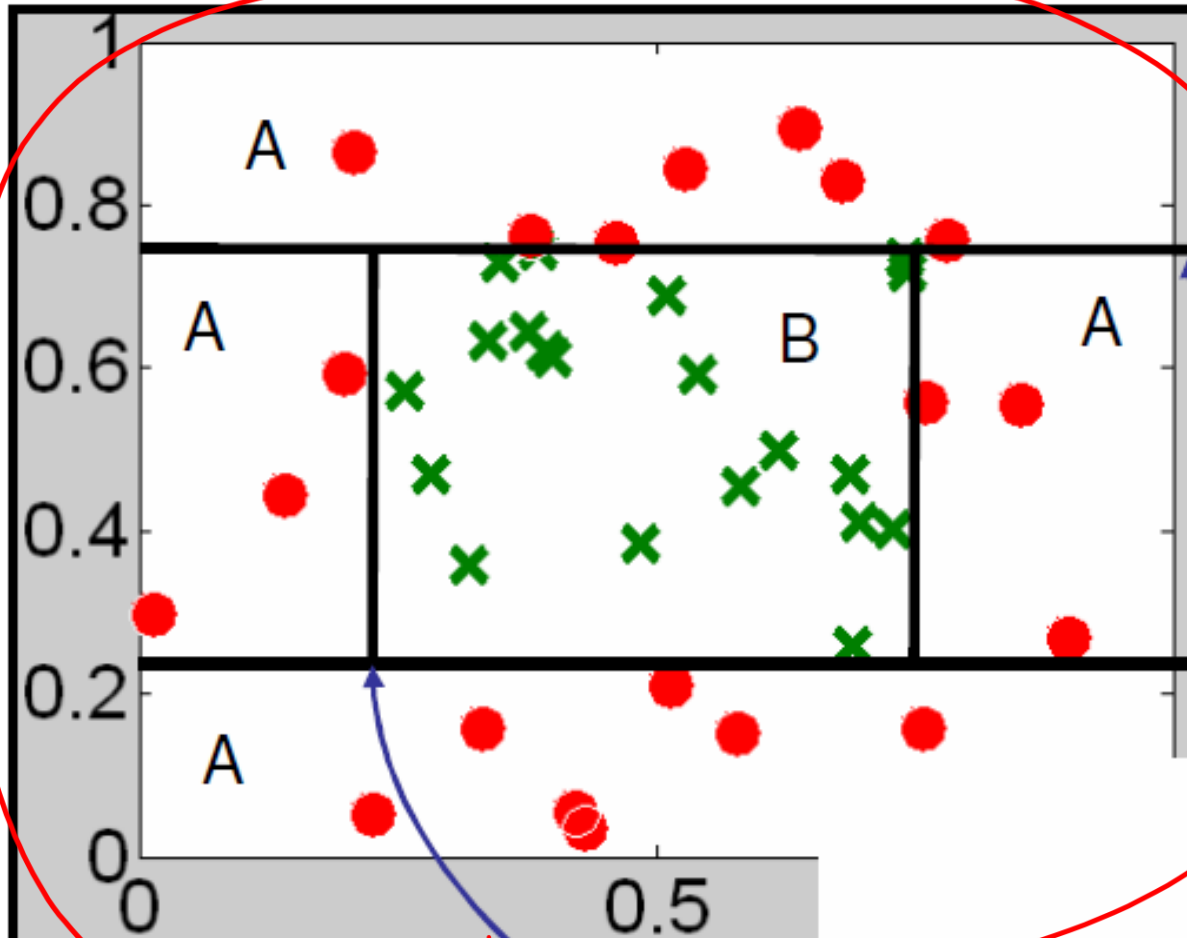


Best X_2 split: 0.0010 0.482
 B 53

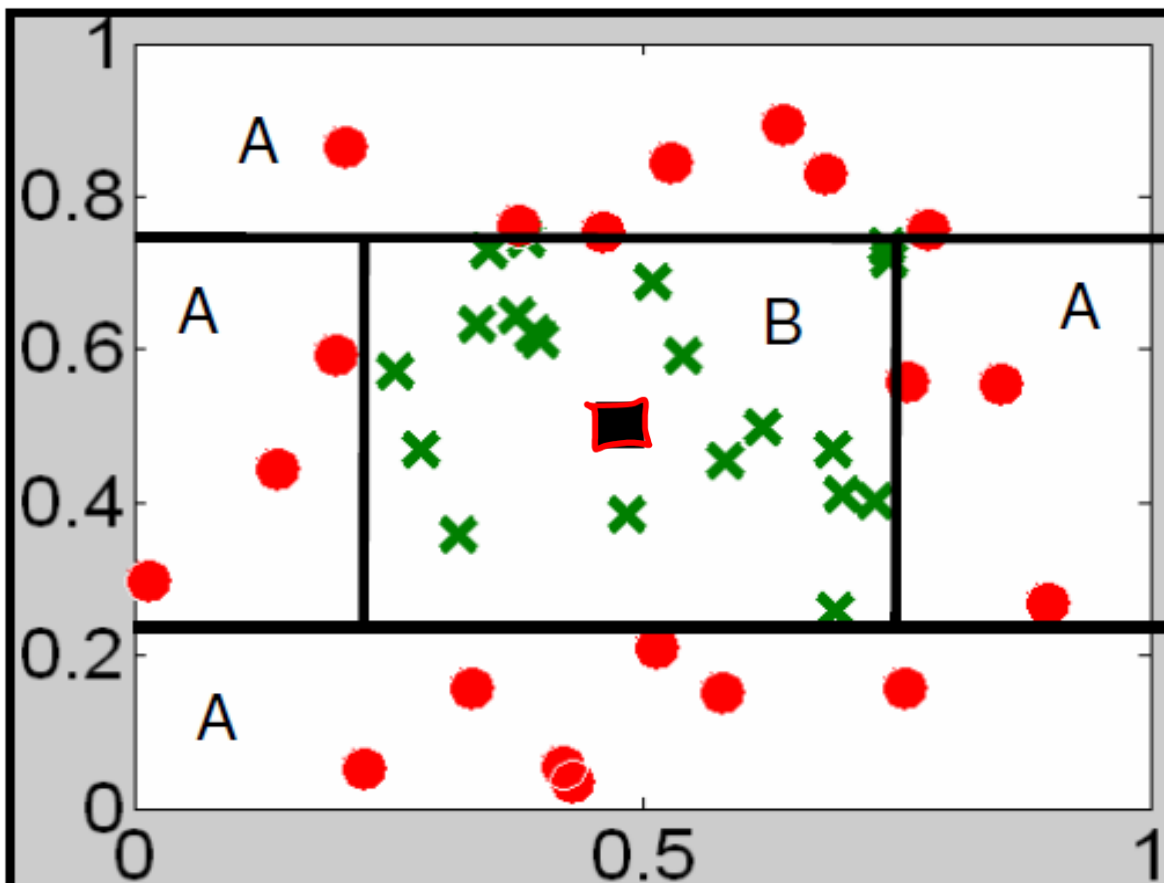
There is no point in splitting this node further since it contains only data from a single class $\rightarrow$ return it as a leaf node with output 'A'



Final decision tree



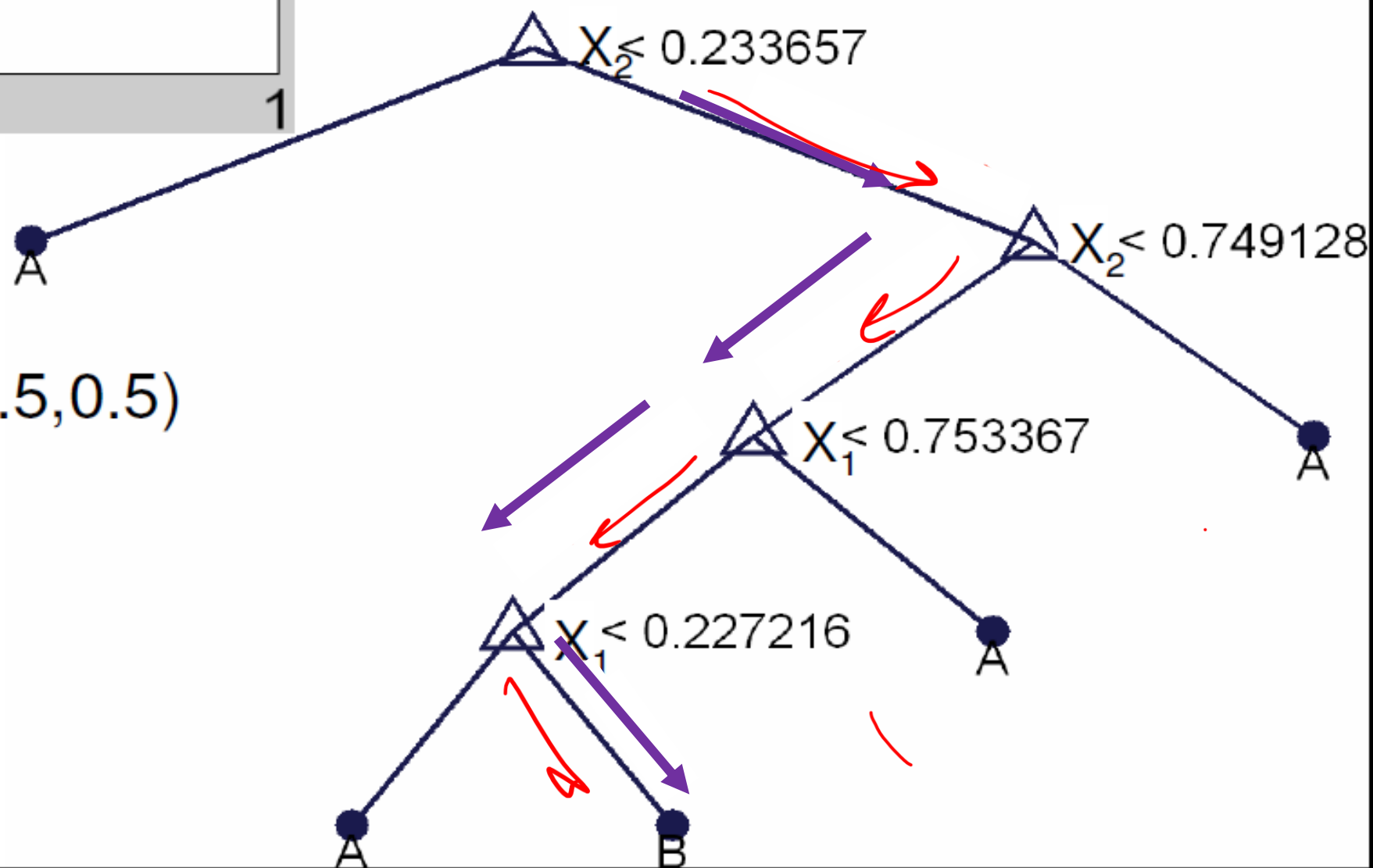
Each of the leaf nodes is pure → contains data from only one class



Final decision tree
 Given an input $(X, Y) \rightarrow$
 Follow the tree down to a
 leaf.

Return corresponding
 output class for this leaf

Example $(X, Y) = (0.5, 0.5)$



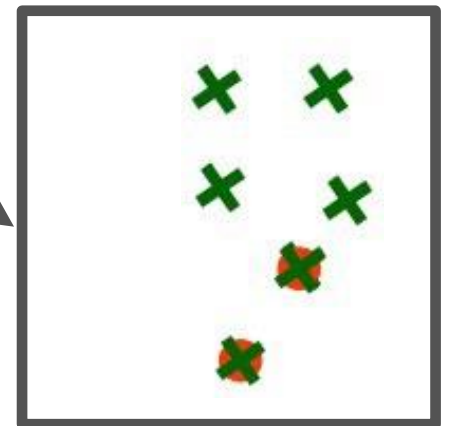
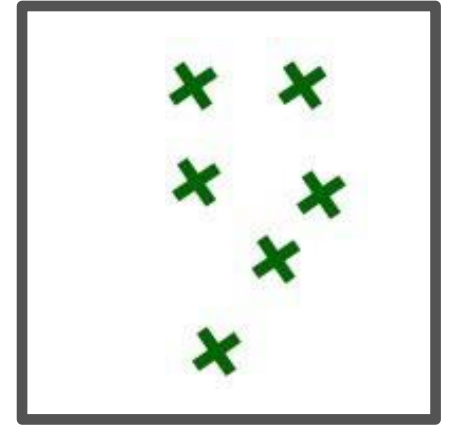
Basic Questions

- How to choose the attribute/value to split on at each level of the tree?
- • When to stop splitting? When should a node be declared a leaf?
- • If a leaf node is impure, how should the class label be assigned?
- If the tree is too large, how can it be pruned?

When to stop splitting? Common strategies:

1. Pure and impure leave nodes

- All points belong to the same class; OR
- All points from one class completely overlap with points from another class (i.e., same attributes)
 - Output majority class as this leaf's label



2. Node contains points fewer than some threshold

5% of training datapoint
↳ Hyperparameter

3. Node purity is higher than some threshold

Pure node: $H(x) = 0$
 $H(x) = 0.2$

4. Further splits provide no improvement in training loss

$$(\text{loss}(T) \leq \text{loss}(T_L) + \text{loss}(T_R))$$

Decision Tree Algorithm (Continuous Attributes)

- LearnTree(X, Y)

Decision Tree is a
computationally expensive
also!

- Input:

- Set X of R training vectors, each containing the values $(x_1, \dots, x_M)$ of M attributes $(X_1, \dots, X_M)$
 - A vector Y of R elements, where y_j = class of the j^{th} datapoint

- If all the datapoints in X have the same class value y

- Return a leaf node that predicts y as output

- If all the datapoints in X have the same attribute value $(x_1, \dots, x_M)$

- Return a leaf node that predicts the majority of the class values in Y as output

- Try all the possible attributes X_j and threshold t and choose the one, j^* , for which $IG(Y|X_j, t)$ is maximum

- X_L, Y_L = set of datapoints for which $x_{j^*} < t$ and corresponding classes

- X_H, Y_H = set of datapoints for which $x_{j^*} \geq t$ and corresponding classes

- Left Child $\leftarrow$ LearnTree(X_L, Y_L)

- Right Child $\leftarrow$ LearnTree(X_H, Y_H)

Decision Tree Algorithm (Discrete Attributes)

- LearnTree(X, Y)
 - Input:
 - Set X of R training vectors, each containing the values $(x_1, \dots, x_M)$ of M attributes $(X_1, \dots, X_M)$
 - A vector Y of R elements, where y_j = class of the j^{th} datapoint
 - If all the datapoints in X have the same class value y
 - Return a leaf node that predicts y as output
 - If all the datapoints in X have the same attribute value $(x_1, \dots, x_M)$
 - Return a leaf node that predicts the majority of the class values in Y as output
 - Try all the possible attributes X_j and choose the one, j^* , for which $IG(Y|X_j)$ is maximum
 - For every possible value v of X_{j^*} :
 - X_v, Y_v = set of datapoints for which $x_{j^*} = v$ and corresponding classes
 - $\text{Child}_v \leftarrow \text{LearnTree}(X_v, Y_v)$

Decision Trees So Far

- Given N observations from training data, each with D attributes X and a class attribute Y , construct a sequence of tests (decision tree) to predict the class attribute Y from the attributes X
- Basic strategy for defining the tests (“when to split”) → maximize the information gain on the training data set at each node of the tree
- Problems (next):
 - Computational issues → How expensive is it to compute the IG
 - The tree will end up being much too big → *pruning*
 - Evaluating the tree on training data is dangerous → *overfitting*

Basic Questions

- How to choose the attribute/value to split on at each level of the tree?
- When to stop splitting? When should a node be declared a leaf?
- If a leaf node is impure, how should the class label be assigned?
- ➔ • If the tree is too large, how can it be pruned?

What will happen if a tree is too large?

Overfitting

High variance

Instability in predicting test data

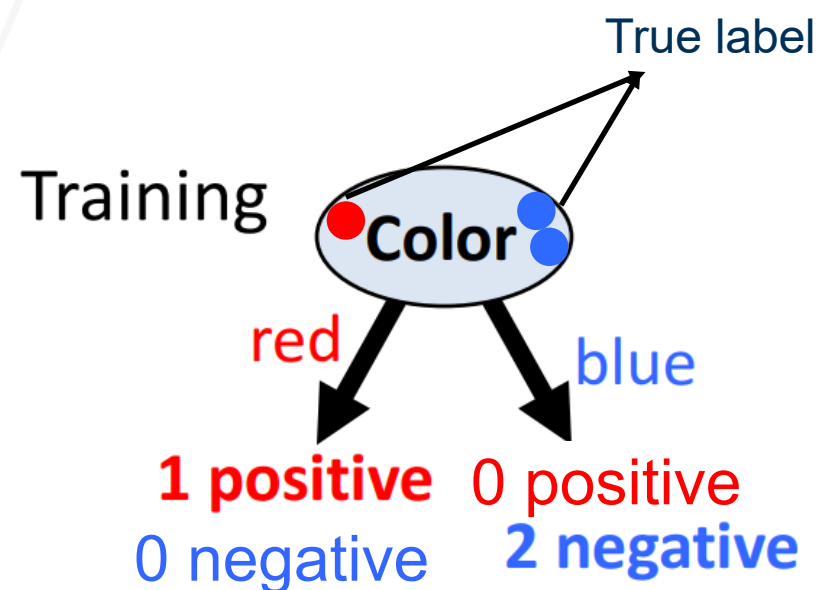
How to avoid overfitting?

- Acquire more training data
- Remove irrelevant attributes (manual process – not always possible)
PCA, forward/backward feature selection
- Grow full tree, then post-prune
- Ensemble learning

Pruning is always done for validation dataset

How to decide to remove it a node using pruning

- Pruning of the decision tree is done by replacing a whole subtree by a leaf node.
- The replacement takes place if a decision rule establishes that the expected error rate in the subtree is greater than in the single leaf.

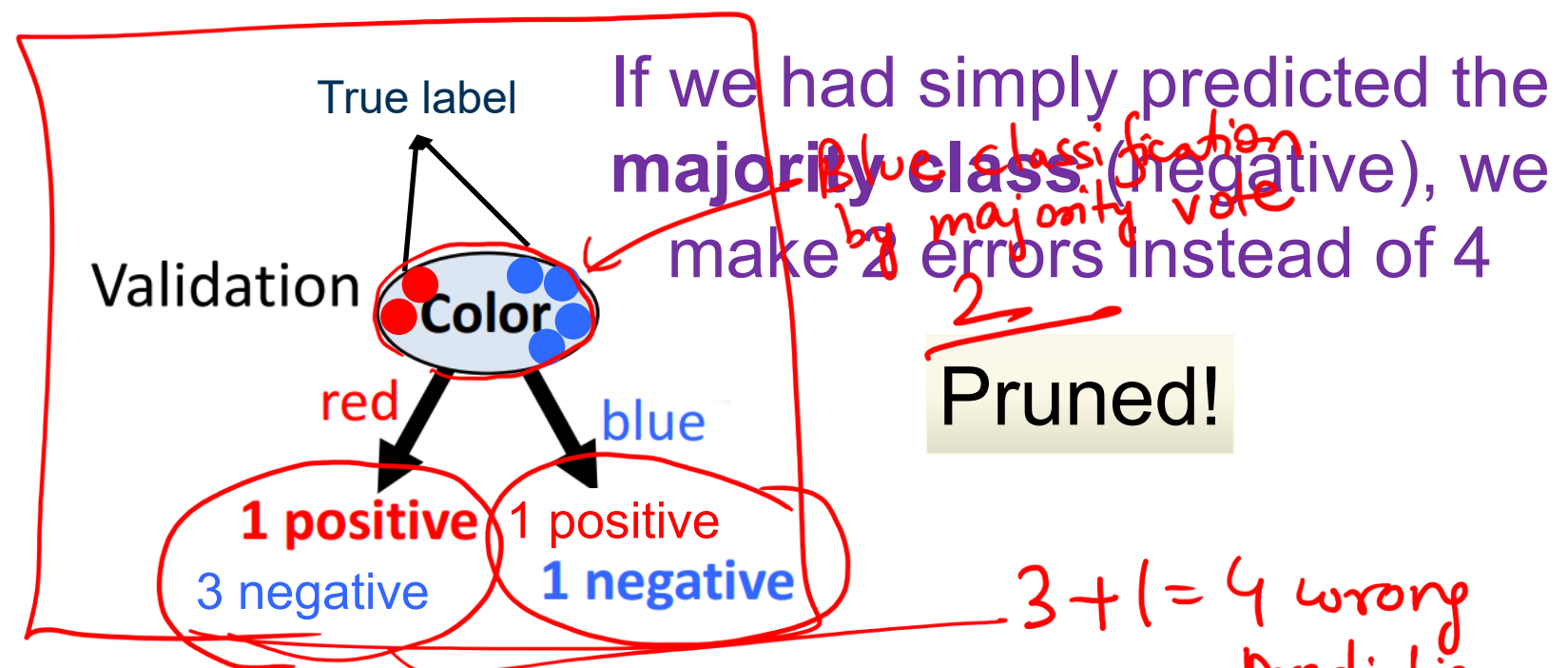


3 training data points

Actual Label: 1 **positive** and 2 **negative**

Predicted Label: 1 **positive** and 2 **negative**

3 correct and 0 incorrect



6 validation data points

Actual label: 2 **positive** and 4 **negative**

Predicted Label: 4 **positive** and 2 **negative**

2 correct and 4 incorrect

Reduced-Error Pruning

Split data into training and validation sets

Grow tree based on training set

Do until further pruning is harmful:

1. Evaluate impact on validation set of pruning each possible node (plus those below it)
2. Greedily remove the node that most improves validation set accuracy